Capstone Design Report

Project Title: Digital Tracking and Accountability Gear (DTAG)

Team #: 1

Team Name: First Response Localization

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Document Change Notice

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Executive Summary

Fire fighters and first responders utilize physical tagging systems to keep track of those on scene. However, this system comes with risks, such as inaccurate data collection, human error, and increased response time. The stakeholders for the project are victims of structure fires and first responding organizations such as police and fire departments. The team is building a digital tag to track and locate members of a first response team when actively working in a potentially dangerous environment. This document outlines the design process, the specifications for the network, and test results on sample networks.

Team Overview

The team is comprised of five members: Rylee Butler, Michael Manning, Jack Torrence, Phil Rodrigues, and Jingyi Yen.

Rylee Butler:

Rylee is the Team Lead for the project. She specializes in control systems and has previous work experience with Android Studio, RFID, root cause analysis on hardware and exterior animation on vehicles. Rylee has written the Research on Existing Technology section in Chapter 1, the Project Requirements, Constraints, and Verification section in Chapter 2, the Prototype Risk Assessment section in Chapter 3, as well as the project risks section in Chapter 8 of the report and hardware build section of Chapter 4.

Michael Manning:

Michael is operating as the assistant team lead for the project. Michael has previous work experience in power and sustainability, dealing with Smart Grid reliability and transmission circuit ratings. He wrote the Problem Discussion section in Chapter 1, the Work Breakdown section in Chapter 2, the Electrical Design and Schematics section in Chapter 3, and the work breakdown summary in Chapter 8 of the report. He also worked on Chapter 5 test results.

Jack Torrence:

Jack is the Technical Lead for the project. He has work experience in signal processing and controls engineering. He specialized in controls engineering. He has written the Introduction and Background section, design/concept section, software design details and the intro paragraph for Chapter 8 of the report. He also worked on Chapter 7 user manual.

Phil Rodrigues:

Phil is operating as the assistant tech lead for this project and is also leading presentation deliverables. His general areas of specialties fall under power distributions and systems as he has worked in internships dealing with lighting, transformers, and large-scale power networks. So far, Phil has contributed on the Social, Environmental, and Global issues and Top-Level Designs and Major Subsystems of this document, as well as the top-level goals in Chapter 8. He also recently contributed to the Mechanical Design and Schematics section of Chapter 3 and all of Chapter 6.

Jingyi Yen:

Jingyi is operating as the Scribe for this project. His general areas of specialties fall under RF antennas as he has worked in internships dealing with RF Optics. So far, Jingyi has contributed to the Market Requirements, Top-Level Preliminary Design and Documentation, Detailed Design, and Electrical Design and Schematics sections. He also ensured all the figures and tables along with the table of contents are formatted correctly. He most recently worked on Software build in Chapter 4.

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Chapter 1: Problem Definition

Introduction and Background

Figure 1 illustrates the current implementation of accountability tagging systems, integral tools employed by firefighters for personnel tracking during emergency operations, encompassing firefighting, search and rescue, and hazardous material incidents. These systems are meticulously crafted to augment both accountability and safety measures. Nevertheless, challenges and issues may arise in their utilization.

A primary challenge pertinent to the physical tagging system is the potential oversight by firefighters to tag in or out during emergency scenarios. This lapse results in a lack of accurate responder status for Fire Chiefs, fostering confusion and an increased risk of potential loss of life.

Furthermore, there exists a vulnerability in the physical tagging system, wherein firefighters or the chief may misplace tags during handovers or storage amid emergency situations. In the event of searching for a specific individual, the Fire Chief must navigate through an array of physical tags, introducing the possibility of misplacement. This scenario amplifies concerns regarding the safety of a firefighter placed in a hazardous situation, contingent upon the retrieval of the correct tag.

Figure 1: Livingston Fire Department Firefighter Tags

Transitioning from physical to digital tagging is a critical step toward improving the safety and efficiency of emergency response operations. The project addresses current challenges in the field and sets the stage for the integration of advanced technologies in emergency management.

Problem Discussion

While firefighters presently rely on a physical tagging system for personnel accountability on scene, this method is susceptible to tracking errors. The primary objective of the physical tagging system is to guarantee the presence and safety of every individual at the incident site. However, instances where a firefighter forgets to tag in before entering a scene pose a potential threat to their safety. Prioritizing safety, the team is exploring a cost-effective digital alternative to mitigate the risk of human error.

The team extensively explored alternative methods to address communication challenges during emergencies. A notable solution found leverages LoRa modulation, specifically Frequency Shift Chirp Modulation [1]. LoRa, a relatively recent technology, employs sinusoids with linearly changing frequencies to transmit messages, utilizing characters. The chirp rate determines the number of characters LoRa can transmit, reaching a maximum of 4,096. Each character corresponds to a specific chirp, and these chirps are transmitted from a LoRa transmitter to a receiver for character decoding [2]. In contrast to traditional methods, LoRa simplifies the decoding process by manipulating the signal and using a Fast Fourier Transform for effective demodulation.

The team identified Meshtastic as a valuable resource for engineering a solution involving LoRa radios and cost-effective components. Meshtastic is a community-driven open-source project that offers various features, including GPS-based location tracking, text messaging, and encrypted communication. Leveraging the capabilities of Meshtastic, the team plans to construct a prototype that addresses the challenges posed by emergency scenarios, employing LoRa technology for reliable tracking.

The team's objective is to address the reliability concerns associated with the physical tagging system, ensuring firefighters can safeguard their personal safety and that of their colleagues. The team aims to develop an affordable device with a streamlined design and user interface to eliminate human error. One proposed avenue for implementation is through long-range infrastructure-free communications (LoRa).

Market Requirements

The project aims to enhance the efficiency and safety of the existing firefighting tag system by providing firefighters with a tool for real-time tracking of their location and status during emergency operations, particularly when entering a structure fire.

Given the critical importance of swift emergency response in firefighting situations, the team is developing an automated tagging system akin to the current physical tagging system. This system, integrated with sufficient resolution, will serve as a real-time tag management tool to ensure the effectiveness and safety of firefighting operations. The proposed solution involves the implementation of wearable Digital Tracking and Accountability Gear (DTAG) on firefighters. The DTAG includes functionality for firefighters to remotely change their tag status from "on

scene" to "interior." Data generated by the DTAG is transmitted to a central LoRa gateway, and if feasible, the system will also transmit GPS coordinates to enhance tracking accuracy.
The primary stakeholders for this project are firefighters, benefiting from improved operational efficiency and safety. Chiefs and Coordinators also stand to gain by having better oversight of their teams and more efficient search capabilities through knowledge of firefighter locations. Additionally, victims of emergency situations benefit from the project's potential to streamline and enhance search and rescue operations. Table 1 provides a summarized overview of the market requirements.

Table 1: Market Requirements

Description of Need	Justification	Requirement / Constraint
Track status of a firefighter's tag	Ensure user can simply see general location of firefighters	Automatically communicate the status of firefighters (off scene, exterior, interior)
Report information back to central console	Easily track firefighters to ensure their safety.	Update location of trackers at least every 5 seconds.
Small, light, and carriable	Do not want the tracking device to inhibit the functions of a firefighter, as they are trying to save lives.	Tracker on firefighter < 1 lb. Central console < 10 lbs. since it would mostly be stationary.
Long battery life	Search and rescue operations take extended periods of time, so the device needs to keep up with that.	Minimum of 8 hours of battery life for the trackers.
Easy to change status	Want firefighters to be able to switch their status with no difficulty, as to not slow down their operations.	Able to switch status quickly while equipped with full standard firefighting gear.
Notify at a change of status	Ensure firefighters are aware system is working.	During a state change, give an indication that state change information has been received, either through LED or speaker on Node.
Function consistently	Firefighters' lives could be on the line, so consistency is crucial.	Consistently track status of tags within 3 meters, at least every 5 seconds, for the entire minimum 8-hour battery life.
Ruggedness	Will be used in emergency environments, which could have lots of rubble. The device will be moving and jostled around a lot as the firefighter search for victims.	The housing for the trackers must be able to withstand 135 N of force (same as if dropping 5 lbs. from 20 ft in the air).
Water and dust resistant	Will be used outside, and disaster relief teams might even go under water or into rubble.	IP68 environmental rating per IEC 60529 (Dust-tight and protected against the effects of continuous immersion in water).

Frequency bands	Must not interfere with national/internationally allocated frequency bands.	Must not interfere with any restricted frequency bands per the Federal Communications Commission (FCC) and the National Telecommunications and Information Administration (NTIA).
(Nice to have) Transmit GPS data of node	Central gateway can display live location of each firefighter, so fire Chief can better coordinate emergency effort.	Visually display GPS coordinates on a digital map.

Social, Environmental, and Global Issues

While the team is primarily focused on the technological aspects of implementing tracking devices using LoRa technology for firefighters, it's crucial to acknowledge and address certain societal concerns associated with these advancements.

One environmental concern involves the responsible disposal of tracking devices utilizing LoRa technology. As these devices often rely on non-degradable battery power sources, sustainability becomes a significant issue. The team aims to minimize environmental impact by exploring sustainable power sources, reducing reliance on disposable batteries, and implementing eco-friendly practices throughout the life cycle of the devices.

Another societal consideration pertains to the potential overreliance on technology, especially in critical operations like rescue scenarios. While the implementation of advanced technologies is beneficial, it is essential to maintain a balance and ensure that traditional methods and skills remain integral. This approach mitigates the risk of societal dependence on technology, emphasizing the importance of ongoing training for rescue personnel.

Affordability is a crucial factor in making these technologies accessible to a wide range of communities. The team recognizes the importance of keeping the cost of these technologies reasonable to enable widespread adoption. If the technologies are prohibitively expensive, many communities may be unable to acquire them, hindering their potential benefits.

Data collection, storage, and distribution present significant challenges, with privacy being a primary concern. The team is committed to safeguarding user data from unauthorized access while maintaining individual privacy. Striking a balance between collecting sufficient data for user safety and protecting user privacy is a key focus.

Ethical considerations also come into play, emphasizing the responsible and ethical use of collected data by authorized users. Preventing misuse, such as unauthorized surveillance, is a critical aspect of the team's approach to ensure that the technology is employed for safety purposes and aligned with ethical standards.

Addressing these societal concerns reflects the team's commitment to a comprehensive and responsible implementation of tracking technologies, considering not only the technological aspects but also the broader impact on the environment, society, and individual privacy.

Research on Existing Technology

As previously mentioned, the most common existing technology is the physical tagging system used by many fire departments. This system includes one red and one green tag that are used to denote where a first responder is on the scene of a fire. The system is simple and widely used, but it requires each firefighter to remember to tag in before each incident. Human error is possible in these scenarios. Despite the risks, the system is cheap and familiar to most first responders, so physical tagging is still the preferred method.

The current physical tagging system demands minimal maintenance and is relatively cost-effective when compared to existing alternative solutions. Many available solutions in the market are characterized by high costs and complexity, often emphasizing dispatch functionalities. An example is the Adashi Firefighter Accountability software, an integrated system encompassing dispatch, engine location information, and firefighter check-ins [3]. Nevertheless, it still necessitates manual check-ins, introducing similar issues to those encountered with the current physical tagging system.

Another example of a high-cost tracking system is NASA POINTER (Precision Outdoor and Indoor Navigation and Tracking for Emergency Responders). This system uses magnetoquasistatic fields to locate first responders in 3D no matter the conditions. POINTER uses small high-precision wearable devices that can direct a responder within centimeters of another responder’s location [4]. Since magnetoquasistatic field can penetrate any material, the technology makes this system ideal for any setting. Although the location precision is better than anything on the market, the system is still in the testing phase. An additional drawback is the cost which makes it impractical for many fire departments with small to medium budgets.

Chapter 2: Preliminary Design Analysis and Concepts

Top Level Design and Major Subsystems

The group's design envisions the incorporation of a LoRa signaling system, featuring a main gateway system capable of communicating with multiple nodes, represented as chips on firefighters. These nodes are designed not only to exchange signals among themselves but also to establish direct communication with the main gateway if within its range. The system optimizes its functionality by requiring only one node to be within range of the main gateway, enabling the rest of the nodes to communicate with each other, thereby extending the operational range of the system for larger areas.

The continuous communication between all subsystems, including the nodes, facilitates a more effective system. The main gateway acts as a central hub, collecting data from the nodes and updating the relative locations on the User Interface (UI). If the main gateway loses track of a node's location or if a specific button is pressed, it will relay the last known location of the node to the UI and mark the node's icon as red.

Figure 2 illustrates a flow chart depicting the functionality of these subsystems, illustrating how the nodes interact with each other in terms of locations and signal processes. This flow chart elucidates the seamless coordination among the nodes, the main gateway, and the UI, ensuring continuous monitoring of a firefighter's precise location and status during an event.

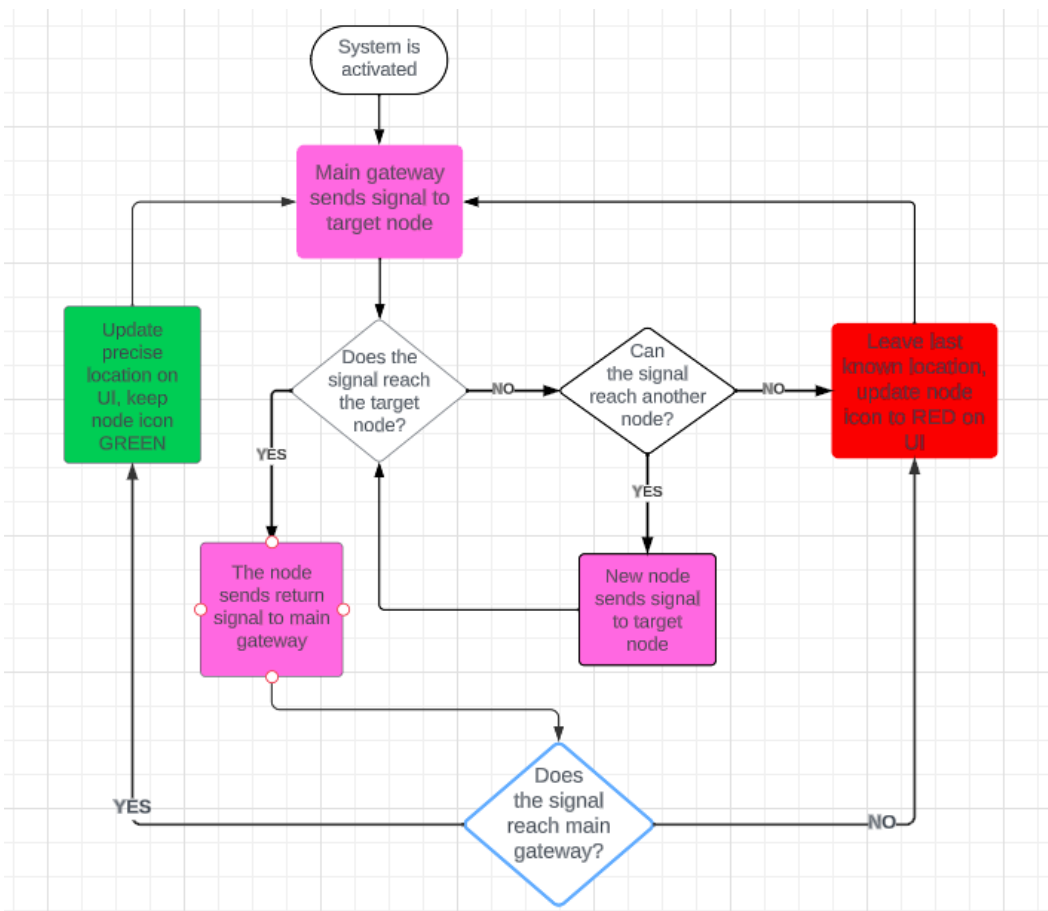

Figure 2: Flowchart of design subsystem processes for GPS display system

As depicted in Figure 3, the team is considering an LED board approach, either independently or in conjunction with a UI display. This subsystem process would operate with its own dedicated operating system. Unlike GPS-based tracking systems, this system follows a manual digital approach, where firefighters press physical buttons on their nodes to update their status. This approach eliminates the necessity of physically submitting a tag, streamlining the process and enabling individuals to enter the scene promptly.

The LED board system provides a tangible and quick way for firefighters to communicate their status, enhancing the efficiency of the overall system. Whether used independently or alongside a UI display, this approach caters to scenarios where rapid updates are crucial, allowing for immediate response without the reliance on more intricate tracking systems.

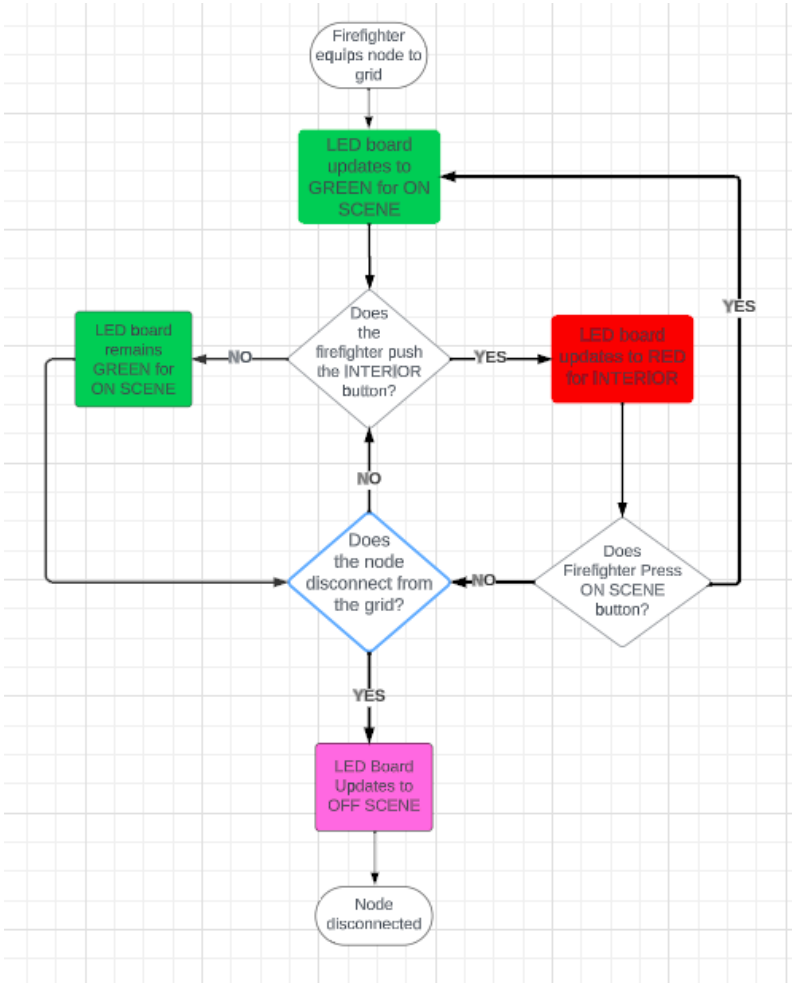

Figure 3: Flowchart of design subsystem processes for LED board system

Design Ideas / Concepts

The team has generated several conceptual approaches aimed at improving upon the current physical tag system, with significant innovations focusing on the user interface (UI) and the precision of firefighter tracking.

In the initial concept, the team proposed a UI that provides the precise location of each tracker relative to the central gateway, as illustrated in Figure 4. This approach offers several notable advantages, including the ability to maintain an attendance record of first responders on the scene by associating tracker IDs with individuals. However, there are potential challenges associated with this concept. The presence of building walls may introduce interference, leading to tracking errors and potential misplacement of responders. In more severe cases of interference, trackers may even fail to transmit signals to the gateway, posing a risk to the accuracy and reliability of the tracking system. The team recognizes the importance of addressing these issues to ensure the effectiveness and robustness of the proposed tracking solution.

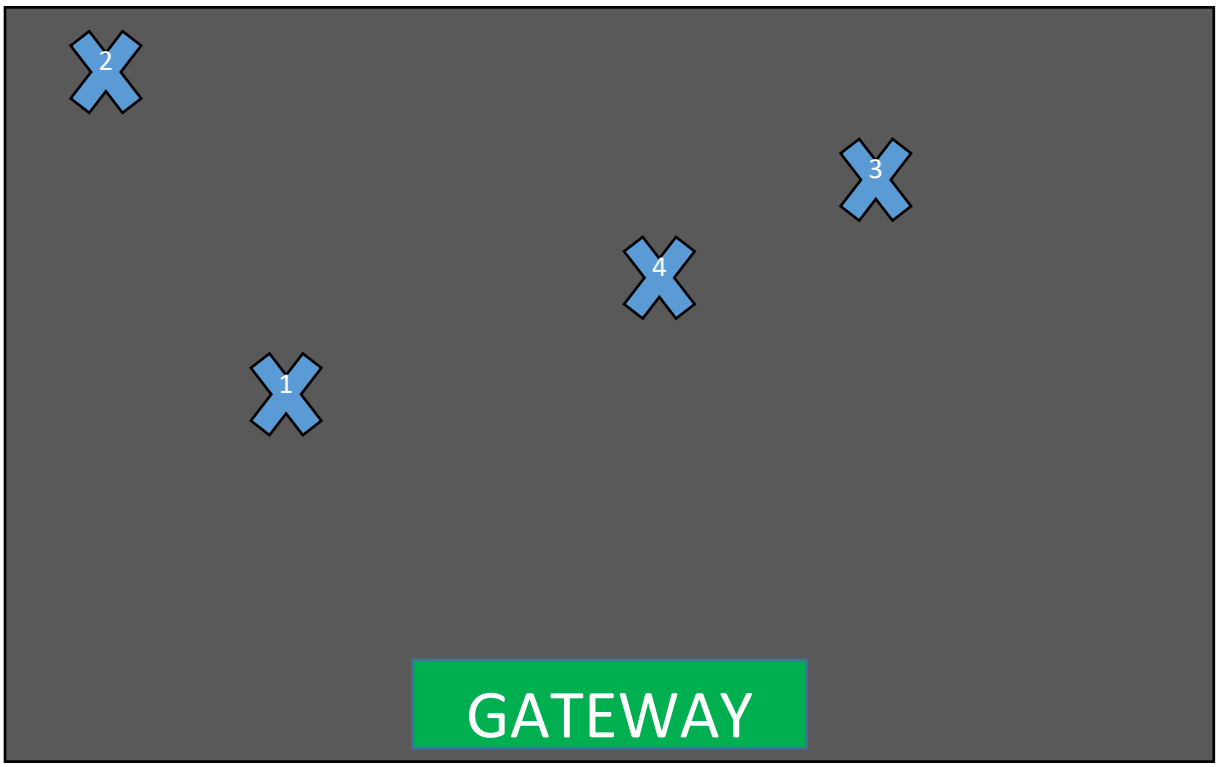

Figure 4: Precise Location UI

In an alternative concept, the team explored the use of an LED board where different colored LEDs signify various states, as depicted in Figure 5. In this concept, one color, likely green, would indicate that a responder is on-site. This could be determined by proximity to the gateway or by the firefighter manually pressing a button on their tracker. Another color, possibly red, would signify that a person has entered a building. This could be achieved if a tracker in the vicinity stops transmitting signals or by the firefighter double-tapping the same button or tapping a different button. If both lights are off, it would indicate that the person is not on-site. This streamlined

approach aims to improve upon the current system of physically checking tags to determine a first responder's presence.

However, a notable drawback of this concept is its lack of scalability due to the required hardware. The dependence on specific LED board components might limit its widespread adoption and implementation. The team acknowledges the importance of considering scalability issues and potential hardware constraints in further developing this concept for optimal effectiveness.

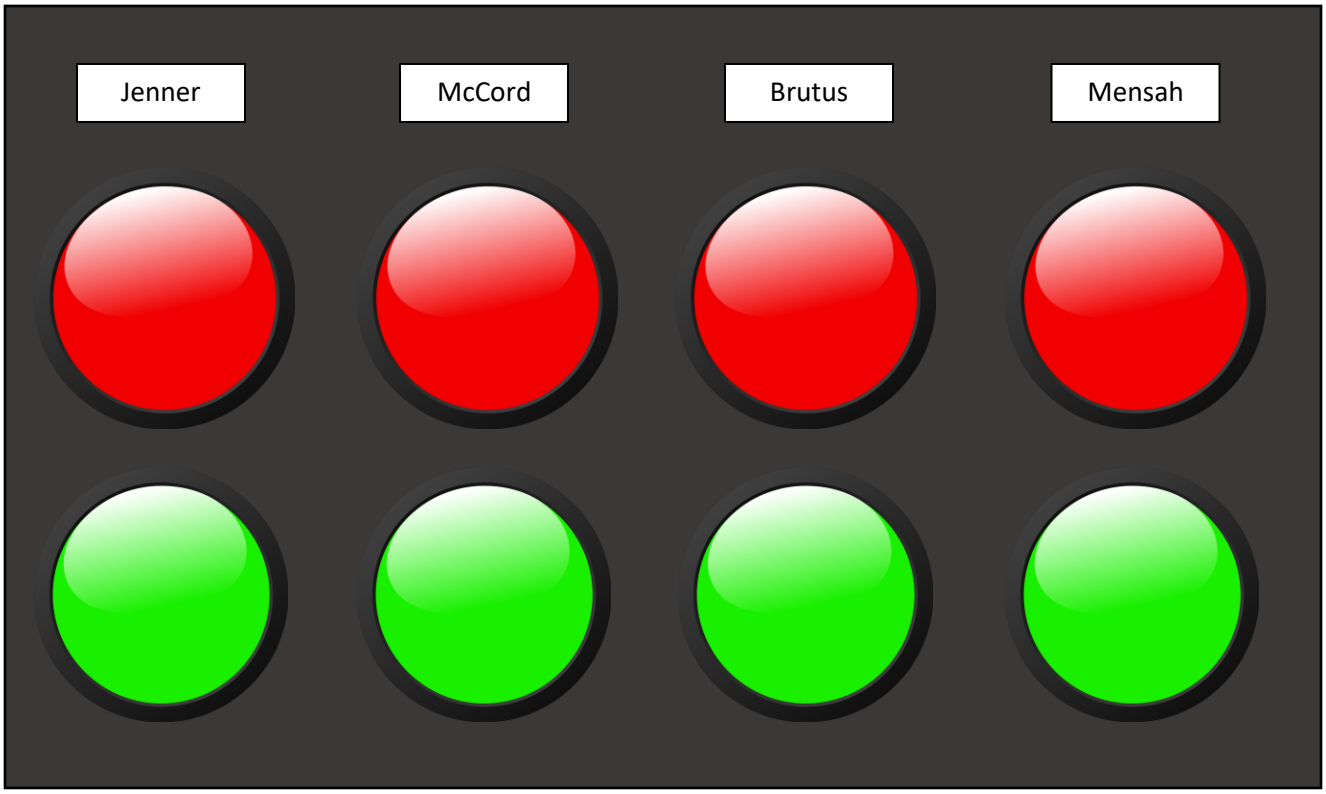

Figure 5: LED Board

After coming up with the two concepts the team made a decision Matrix displayed in below to help decide what concept to pursue. This matrix compared the current system of using physical tags, tracking with precise location, and using an LED board. Some of the criteria used in the decision matrix were scalability, efficiency and simplicity.

Table 2: Decision Matrix

	Physical Tags	**Tracking Board**	**LED Board**
Identifies who is on site	✓	✓	✓
Efficient	X	✓	✓
Simple	✓	✓	✓

Scalable	✓	✓	X
Gives location (Vague)	X	✓	✓
Works if tracker signal lost	X	X	✓

Upon further deliberation and amalgamation of concepts, the team arrived at a final idea that capitalizes on the strengths of both previously described concepts. This ultimate concept, illustrated in Figure 6, merges the user interface (UI) from the first concept with the broader tracking approach from the second concept.

In this final concept, the UI provides individuals' information along with one of three states: interior, exterior, or offsite. To achieve this, firefighters would carry a device equipped with input devices, such as buttons, allowing them to update their relative position. When arriving on-site, a user would press a designated button, ideally receiving a response from the system to confirm the successful transmission of their message. If the user needs to enter a building, a separate button press would update their position accordingly. The UI for this system maintains the color scheme of accountability tags for seamless transitions between systems. Depending on the available resolution, additional location information may be displayed on the UI.

This integrated concept aims to offer a comprehensive solution, leveraging the benefits of both precise location tracking and a simplified LED board-like approach. The UI provides essential information in real-time, enhancing accountability and situational awareness for first responders.

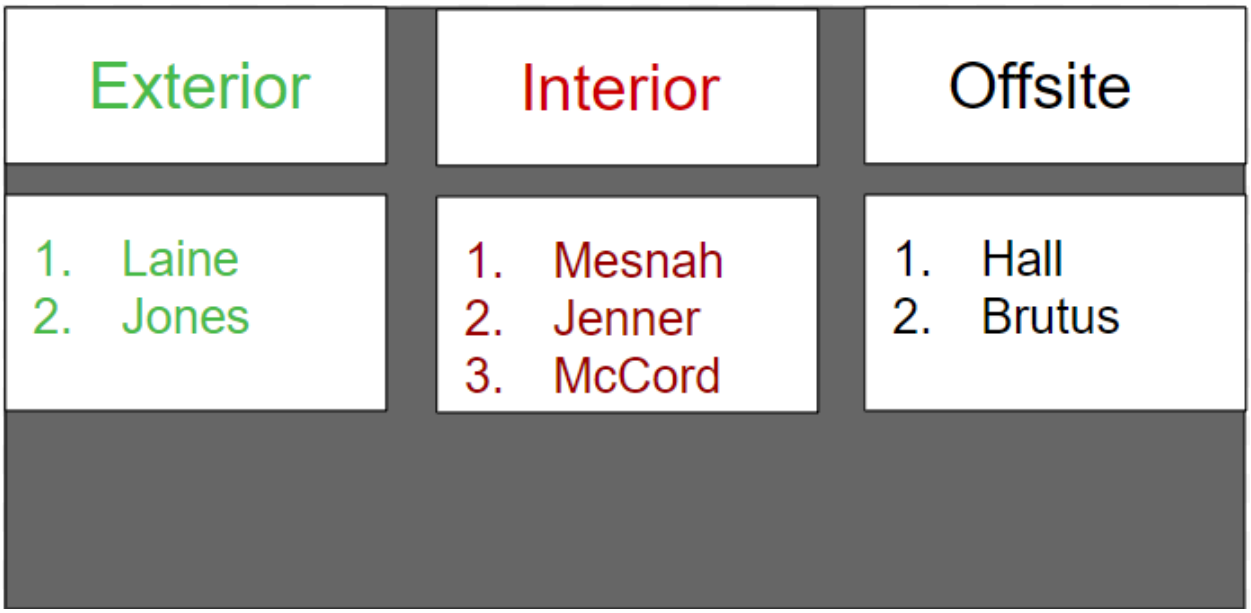

Figure 6: Status Indicator UI

Project Requirements, Constraints, and Verification

Upon further deliberation and amalgamation of concepts, the team arrived at a final idea that capitalizes on the strengths of both previously described concepts. This ultimate concept, illustrated in Figure 6, merges the user interface (UI) from the first concept with the broader tracking approach from the second concept.

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This integrated concept aims to offer a comprehensive solution, leveraging the benefits of both precise location tracking and a simplified LED board-like approach. The UI provides essential information in real-time, enhancing accountability and situational awareness for first responders.

Table 3: LoRa Mesh System Technical Requirements and Verification

Description of Need / Design Parameter	**Justification**	**Requirement / Constraint**	**Verification Methodology**	**Comments**
Light Weight	Responders will carry a node on their person	Node <= 1 lb	Scale	N/A
Limited Frequency Band	Frequency bands allowed without filing with FCC: 900 MHz, 2.4 GHz, 5.8 GHz	915 MHz band	N/A	LoRa has an assigned frequency of 915 MHz
Range	Coverage for area of scene	>= 500 ft	Range test module (provided by Meshtastic)	N/A
Noise Filtering	Mitigate noise from surrounding environment	Spreading factor > 9	Analysis and empirical testing	Greater spreading factor = more range, better noise suppression
Validation for State Changes	Firefighters won’t immediately trust a digital system	During a state change, give indication that state change has been received	Empirical testing	May be visual or auditory

Upon completing the prototype, the team will undertake a series of validation tests to ensure the solution aligns with the specified requirements and constraints. Of particular significance are the range and noise filtering tests, which are pivotal in verifying the success of the system.

To assess the range of the LoRa mesh system, the team will employ the range test module provided by Meshtastic [5]. This module facilitates the measurement of the system's effective

communication distance. During this test, the receiver will capture signals transmitted by the node, including its location information. The recorded data will be saved in a .csv file for further analysis.

By conducting these validation tests, the team aims to assess the system's performance in real-world scenarios, particularly in terms of its operational range and its ability to filter out noise. These tests serve as critical measures to validate the functionality and reliability of the developed solution.

WBS, Gantt Chart, and Schedule

The team detailed the work breakdown for the remainder of our project in a Gantt Chart to remain on schedule and ahead of any issues that might arise throughout the project's development.

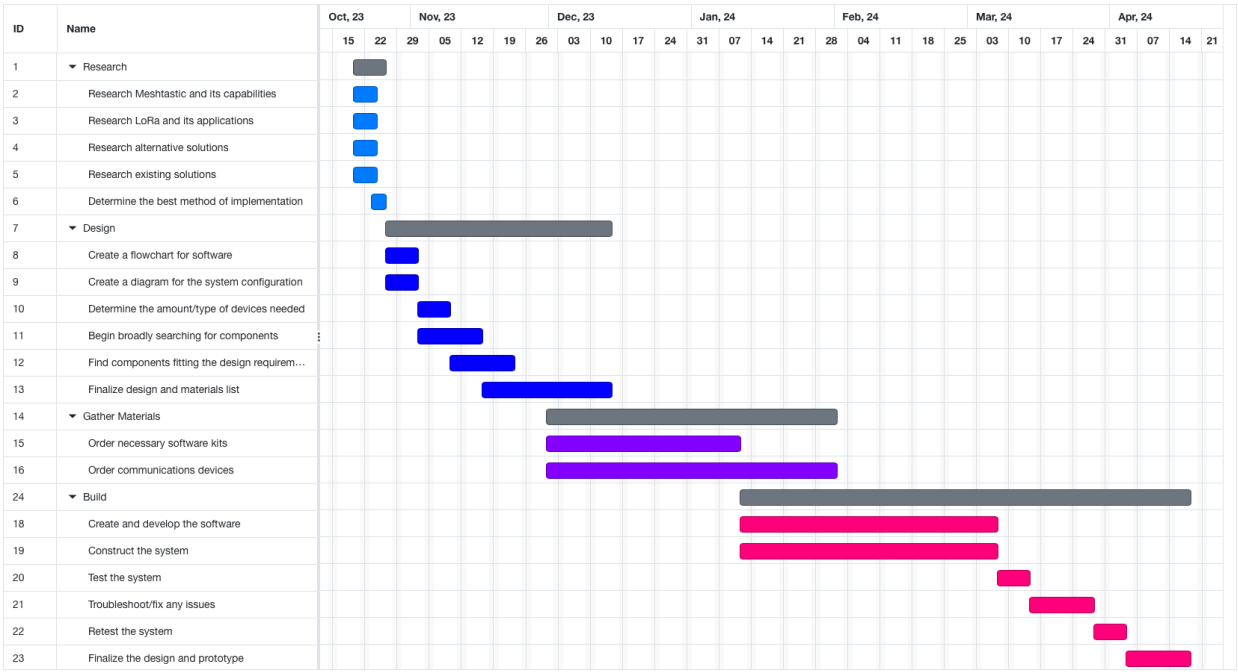


Figure 7: Gantt Chart for Remaining Work Breakdown

The Gantt Chart in Figure 7 shows the work breakdown for the remainder of the project through April 2024. The chart splits the remaining work into four sections: research, design, gather materials, and build.

Each of the remaining tasks in the Gantt Chart were allocated to team members based on level of expertise and comfort level. The Gantt Chart software provides a filtered view of the project to easily view each member's role in the project.

The Gantt Chart can be found in the following link to take a closer look at the work breakdown for the DTAG team.

https://www.onlinegantt.com/#/gantt/cloud

Top Level Preliminary Design and Documentation

In the design of a tracking system, the range of tracking devices is a critical consideration. A proposed solution involves leveraging the Meshtastic system [6], as illustrated in Figure 8. In this configuration, instead of each individual node communicating solely with the central gateway,

each node establishes communication with other nodes within the mesh network. The information gathered by these interconnected nodes is then funneled back to the central gateway. This approach has the potential to significantly extend the overall range of the mesh network, as it is not solely reliant on the individual range of the central gateway.

By adopting a mesh network topology, the tracking system gains the advantage of improved coverage and range through the collaborative communication of nodes. This decentralized communication model enhances the system's robustness and ensures a more extensive and resilient tracking capability, particularly in scenarios where individual nodes may be at a distance from the central gateway. The effectiveness of this solution lies in its ability to harness the collective communication capabilities of all nodes within the mesh, providing a more comprehensive and adaptable tracking infrastructure.

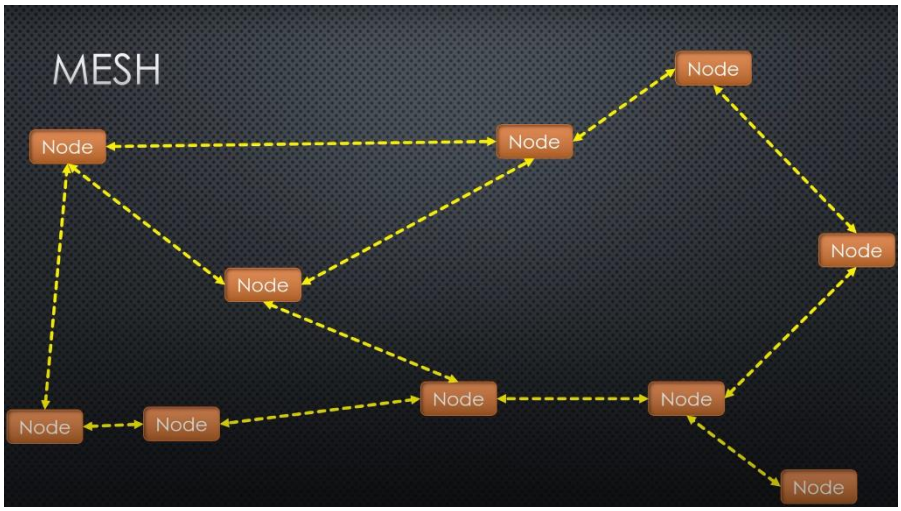


Figure 8: Block diagram of nodes communicating with each other [6]

An additional advantage of employing the mesh approach is the capacity for communication between nodes even in the absence of a direct line of sight. This capability is illustrated in Figure 9, where nodes within the mesh network establish communication pathways irrespective of potential obstacles. It is noteworthy that the central gateway, serving as a pivotal component, is also integrated into the mesh as a node.

This feature ensures that nodes within the tracking system can effectively relay information to one another, creating a resilient network that can navigate challenging environmental conditions. The ability to establish communication links between nodes regardless of obstacles enhances the system's adaptability and robustness, contributing to a more reliable and versatile tracking

infrastructure. This approach mitigates potential challenges posed by physical barriers, further solidifying the mesh system's suitability for diverse operational environments.

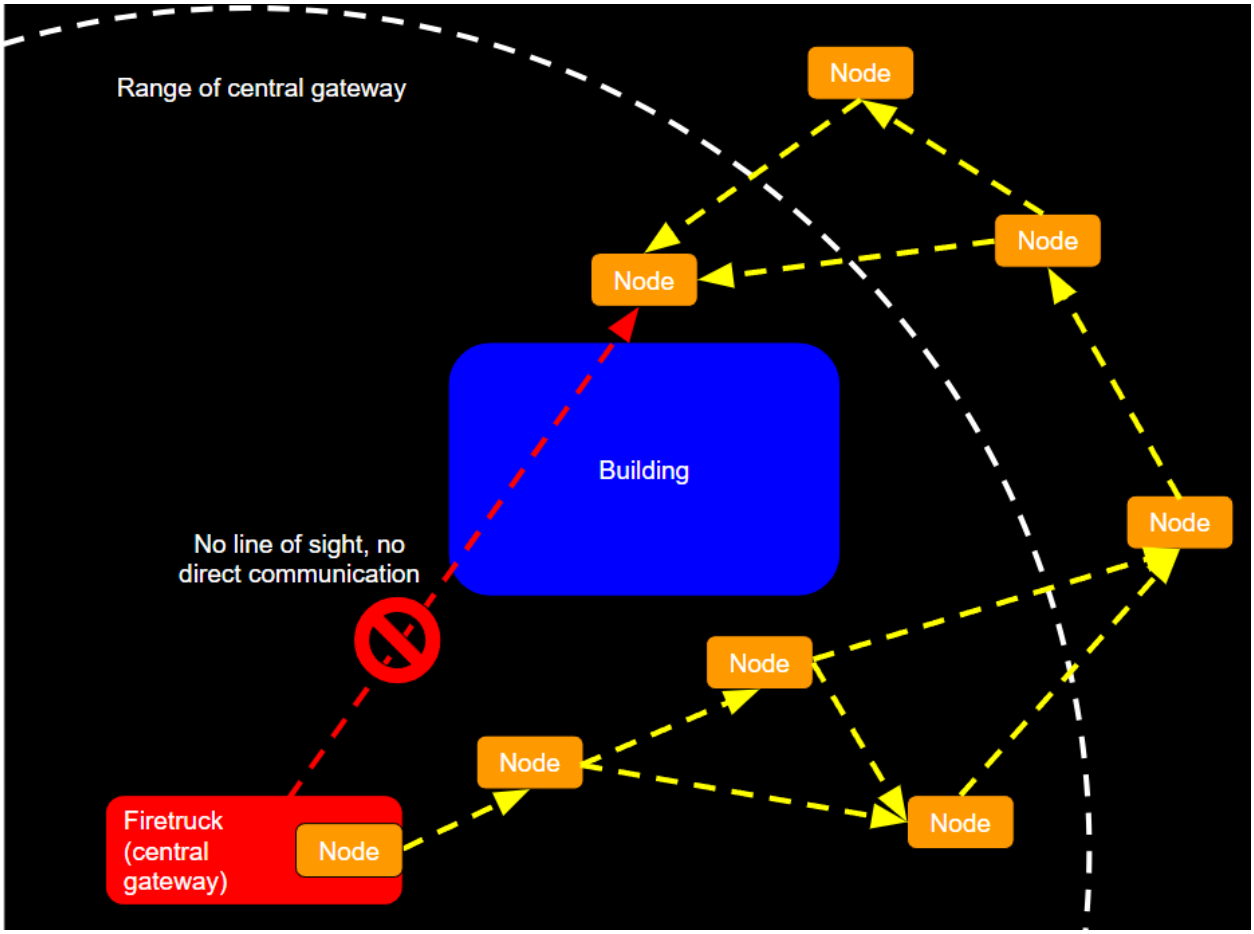


Figure 9: Extended range and working around line of sight with Meshtastic system

In Table 4 is the list of materials needed for this Meshtastic system, with the main subsystems being the central gateway, and the individual nodes for each firefighter. Figure 10 is the LILYGO TTGO ESP32 communication board selected to be used as the nodes.

Table 4: List of Materials

Subsystem	Component	Quantity
Central Gateway	Meshtastic Web Client	1
Node/MCU	LILYGO TTGO LoRa32 ESP32 board	1 for each first responder + 1 for the central gateway
Push Button	QTEATAK 6 x 6mm x 5mm Momentary Tactile Tact Push Button Switch	2 for each node to indicate exterior / interior status
Pull Down Resistor	10 kΩ resistor	2 for each node, connected to push button
Speaker	Round Micro Speaker, 28mm 2W 8 Ω	1 per node

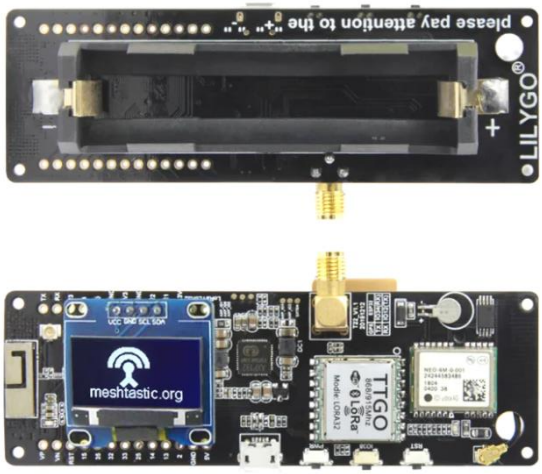


Figure 10: LILYGO TTGO ESP32 T-Beam V1.2 [7]

Chapter 3: Detailed Design

System Diagram

The team has decided to implement the GPS coordinates and status user interface (UI) concepts outlined in Figure 2, Figure 4, and Figure 6. Leveraging the Meshtastic web client, which already displays GPS coordinates on a map similar to Figure 4, the team plans to implement the status UI, as depicted in Figure 6, using Python. This approach aligns with firefighters' familiarity with various web and mobile applications for personnel allocation, making the proposed system seamlessly integrate into their existing workflow.

Each node within the system will be equipped with the same ESP32 chipset, featuring an MCU capable of LoRa communication, an antenna, a GPS module, push buttons for status changes, a speaker for user feedback, and Bluetooth capability. The Bluetooth functionality is specifically intended for the central gateway's designated node to pair with a laptop running the Meshtastic web client. This enables the transmission of GPS coordinates and status updates for each node—exterior, interior, or offsite. The Meshtastic GPS display and the Python-based status UI will utilize this information for real-time monitoring.

For a more detailed explanation of the Python GUI and software design, please refer to the upcoming Software Design Details section. Figure 11 provides a functional block diagram illustrating the communication between various components of the central gateway and nodes in

the system. This interconnected design ensures efficient data flow and communication among the different elements of the tracking system.

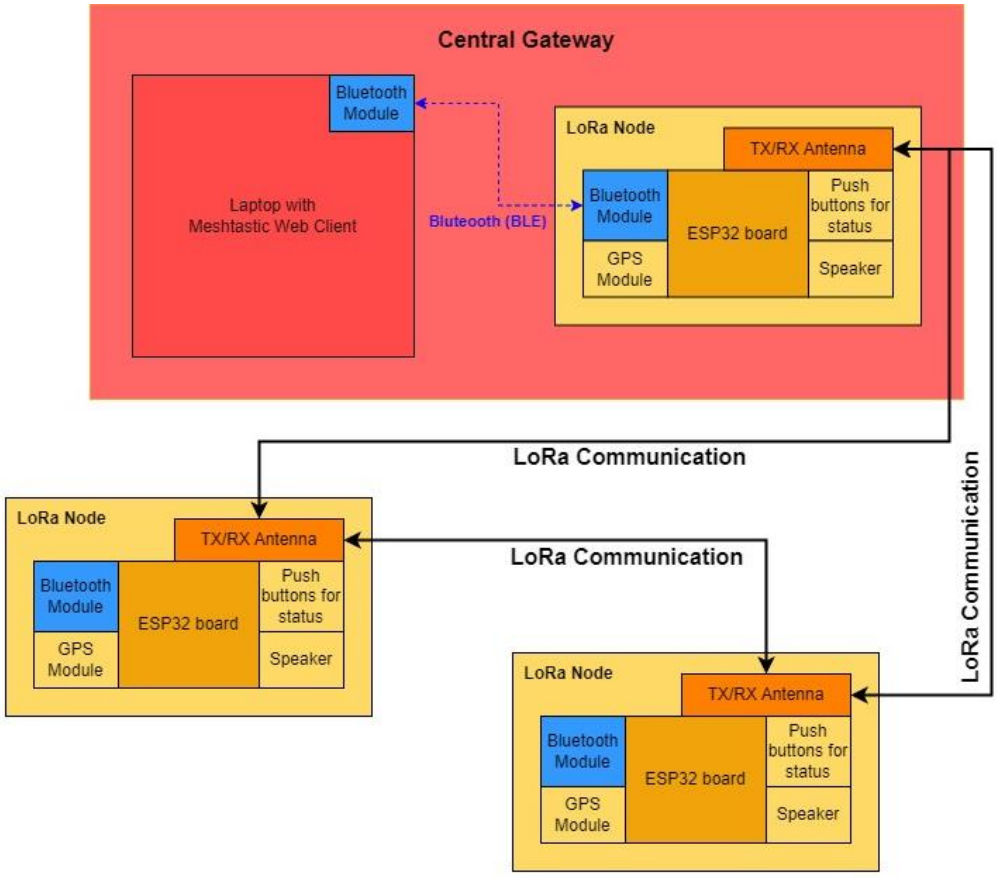


Figure 11: Functional Block Diagram of Central Gateway and LoRa Nodes

Electrical Design and Schematics

The team developed a final design for our DTAG tracking system and began to buy parts for the prototype. The tables and figures in this section show the main components and how they meet the critical requirements outlined by our problem definition.

Table 5 shows the specifications for the LILYGO TTGO ESP32 module. These modules are for the nodes and central gateway.

Table 5: Specification Table for the ESP32 MCU Board

Specification	**ESP32 Functionality**	**Meets Requirements?**
Frequency Band	915 MHz	Yes
GPS Tracking	Included	Yes
Bluetooth	Included	Yes
Meshtastic Capabilities	Pre-Installed	Yes

The ESP32 board meets the basic requirements to create a functioning tracking system. The ESP32 makes it easier for the team to program the system since it is already integrated with Meshtastic.

The ESP32 provides built-in Bluetooth connectivity and GPS tracking for seamless connectivity between devices without the need for additional programming or components.

To supply power, each ESP32 requires one NCR18650GA battery. Table 6 displays the specifications for the battery.

Table 6: Specification Table for the NCR18650GA Battery

Specification	**NCR18650GA Functionality**	**Meets Requirements?**
Nominal Capacity	3400mAh	N/A
Discharge Current	10A	N/A
Nominal Voltage	3.6V	N/A
Battery Life	Rechargeable	Yes
Charging Time	~4 hours	Yes

The rechargeable battery chosen for each ESP32 board offers a lifespan of up to 300 charging cycles, maintaining 70% capacity. With a relatively short charging time of 4 hours, the battery is conveniently rechargeable overnight or during alternative battery usage. Although specific details about the battery life per cycle are not explicitly provided, noteworthy cases have demonstrated batteries lasting up to 28 hours in systems with 7 devices utilizing Bluetooth and GPS services. Given the constraints of the DTAG tracking system, this battery solution is deemed more than adequate for sustained tracking purposes.

The LILYGO TTGO ESP32 package comes pre-soldered, but the team plans to enhance it by adding a large push button to facilitate status changes, even when firefighters are wearing gloves. Additionally, an external speaker will be incorporated to provide audio feedback indicating a status change. Figure 12 presents the pin diagram of the LILYGO TTGO ESP32 chipset, and Figure 13 depicts the electrical schematic of the ESP32 chipset implemented within the team's design. In Figure 13, the highlighted sections denote components not included in the pre-soldered LILYGO TTGO ESP32 chipset. It's essential to note that the specific components for the speaker, large push button, and resistors have yet to be selected.

This battery and component configuration, when implemented in the tracking system, ensures reliable and user-friendly functionality for firefighters in the field. The design accounts for usability concerns, such as operating with gloves, and incorporates features like audio feedback to enhance the overall user experience. As the team proceeds with component selection, attention to these details will contribute to the success of the DTAG tracking system.

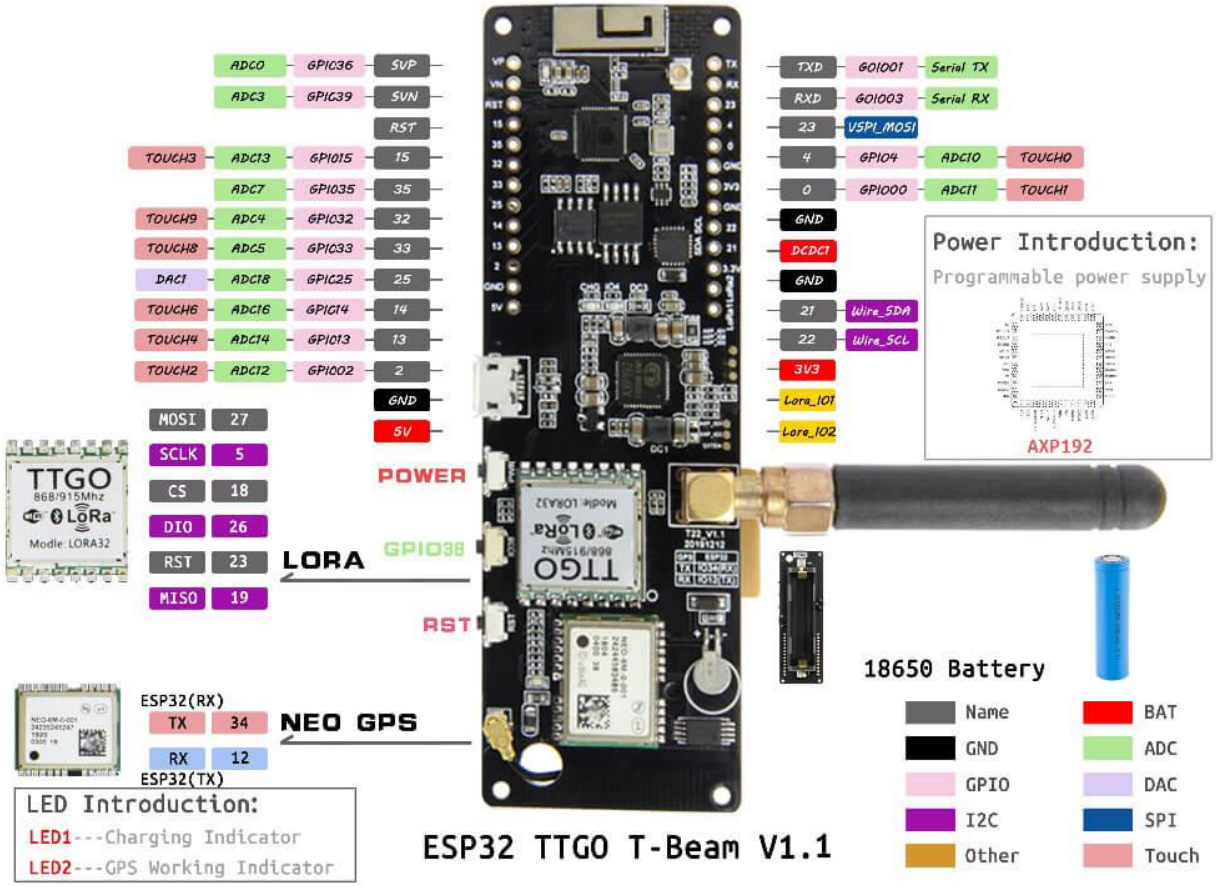


Figure 12: Pin diagram of LILYGO TTGO ESP32 chipset [7]

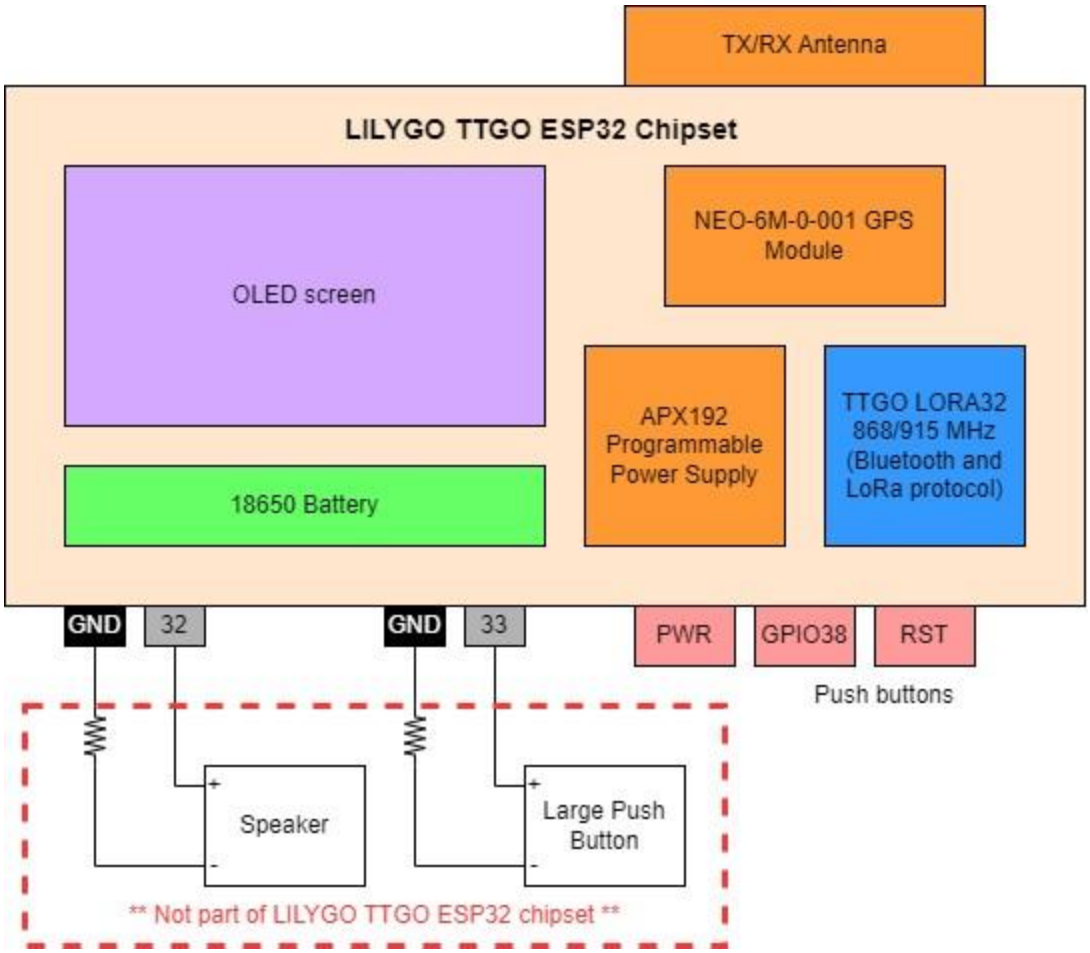


Figure 13: Electrical schematic ESP32 chipset with external speakers and large push button

The team bought the ESP32 which is most of the cost of the total project. However, there are still components that factor into the final design that the team needs to buy. The theoretical bill of materials and the theoretical and actual costs are shown in Table 7.

Table 7: Theoretical Cost of the Materials

Item (# of Units)	**Pre-Shipping Cost**	**Shipping Cost**	**Theoretical Cost**	**Actual Cost**
ESP32 MCU (5)	$224.85	$0.00	$224.85	$89.94
NCR18650GA (2)	$12.99	$0.00	$12.99	$12.99
Charger (1)	$7.99	$0.00	$7.99	$7.99
Speakers (4)	$14.76	$6.99	$21.75	$21.75
Resistors (10)	$1.00	$6.99	$7.99	$0.00
Push Button (4)	$8.90	$6.99	$15.89	$0.00
Totals			**$291.46**	**$132.67**

The resistors and push buttons are two components that the team has and plans to implement into the design by incorporating them into the status changing system. First, the team's focus is on

troubleshooting the ESP32 boards and getting them to connect with each other using the built-in Bluetooth and GPS functionality.

Software Design Details

The software for this project is structured into four distinct parts, predominantly written in Python, and utilizing the Meshtastic API [8]. The initial segment is a start-up program, initiated upon user request at the central gateway. Its primary function is to obtain a count of nodes within the vicinity and retrieve the corresponding nodal IDs. This information is then matched to users, and a graphical user interface (GUI) is generated. All users with nodes in the vicinity initially commence in the exterior state. All other users are set to the exterior state. Subsequently, the standby state is initiated for all nodes not directly connected to the central gateway. The central gateway will, at fixed intervals, check for new nodes in the vicinity.

The individual nodes will produce a sound at the speaker when turned on. The user of the node will know the node in standby at this point. Nodes will remain in standby until turned off, signified by the OLED screen turning off. Nodes in standby will be able to run two functions with different priority and starting criteria. The State Change Request function is of higher priority and is triggered when the node recognizes one of a set of button patterns pressed. Figure 14 illustrates the flowchart for the State Change Request function. The message associated with a state change request includes information on the requesting node and the state change that corresponds to the button pattern performed. The software is set up in this way so if a node is damaged or turned off, the last confirmed state remains where the users name is at the central gateway.

The flowchart provides a visual representation of the sequence of actions involved in processing a state change request. As the operational function serves a critical role in managing node states, this

comprehensive flowchart ensures clarity and understanding of the software's behavior during state change requests.

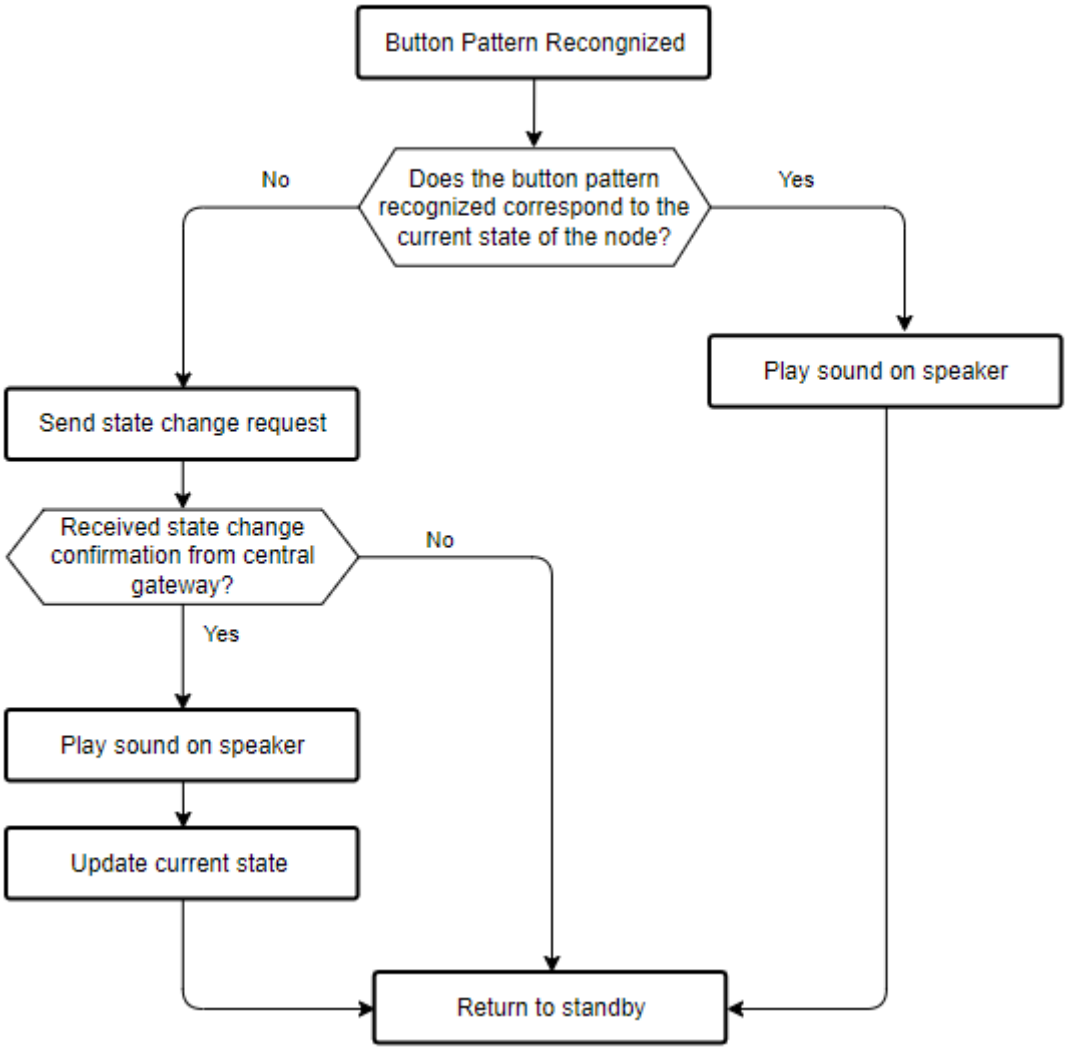


Figure 14: State Change Request Function Flowchart

Continuing with the software architecture, the third part of the system operates at the central gateway. When the node connected to the central gateway receives a state change request, it processes the request by acknowledging its reception and matching the nodal ID to the corresponding user. The user's name is then moved to the GUI column associated with the requested state. This ensures a visual representation of the current state of each user in the graphical user interface.

The final part of the software, exclusive to the central gateway, runs continuously from the end of setup until the device is powered off. This component functions every five seconds, utilizing GPS to retrieve the location of all nodes in an operating state. The goal is to update the location of each node as GPS signals are received, facilitated through the Meshtastic Web Client. If no

signal is detected, the node's icon undergoes a visual change, likely turning red, signaling to the user that the location information is outdated.

This systematic approach to software development ensures efficient communication and coordination among nodes while providing real-time updates on the location and state of each user. The prioritization of state change requests, mesh communication, and continuous GPS updates reflects a thoughtful design that aligns with the project's objectives and user requirements.

Mechanical Design and Schematics

In the mechanical aspect of the team's tracking device, the casing plays a crucial role in the overall design. The casing is of paramount importance as it serves to shield the Microcontroller Unit (MCU) from potential hazards such as dust, debris, and heat, which are prevalent in the challenging environments that firefighters often encounter during emergency responses.

The team has identified existing casing designs for the Meshtastic T-Beam available online, and Figure 15 provides a visual representation of these designs. The intention is to leverage these existing designs and modify them to align with the specific requirements and needs of firefighters.

The adaptation of these casing designs ensures a protective enclosure for the tracking device, safeguarding the internal components from external elements that could potentially compromise the device's functionality. By customizing the designs to suit the unique demands of firefighting scenarios, the team can enhance the durability, resilience, and overall effectiveness of the tracking device in real-world operational conditions. This approach not only expedites the mechanical design process but also ensures that the final product is tailored to meet the specific challenges faced by firefighters during emergency situations.

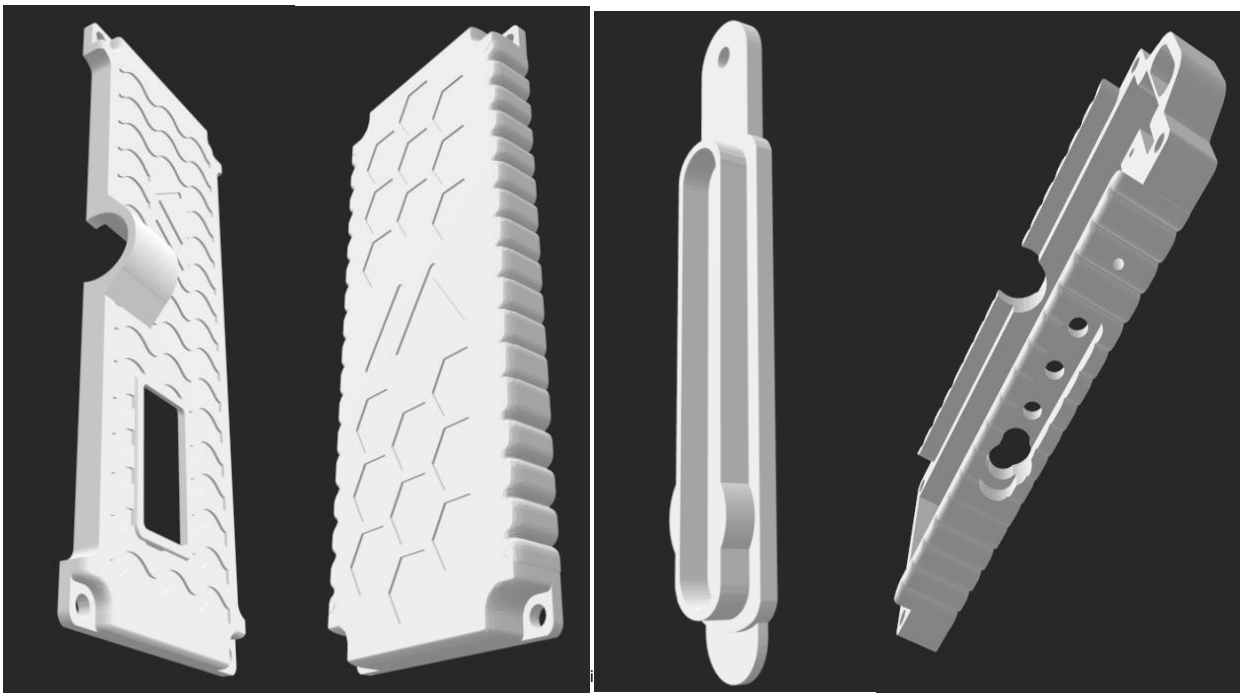


Figure 15: Casing Design of Front Cover, Back Cover, Dust Cover, and Outer Frame [9]

The incorporation of an element-proof casing, as demonstrated in Figure 15, is pivotal for ensuring the durability and resilience of the Microcontroller Unit (MCU) in challenging firefighting environments. This casing design is crafted to shield the MCU from external factors such as dust, debris, and other elements that could potentially compromise the device's critical electronics. The robust casing is essential for maintaining the operational integrity of the tracking device, safeguarding its internal components from environmental stressors encountered during emergency responses.

In addition to the element-proof casing, Figure 16 highlights the importance of designing a gasket piece to seal the opening created for the antenna. This gasket serves as an additional protective measure, preventing the ingress of dust and other elements through the antenna opening. The gasket acts as a barrier, ensuring that the internal components remain secure and unaffected by external environmental factors.

Together, the element-proof casing and the gasket contribute to the overall ruggedness and reliability of the tracking device. By addressing potential points of vulnerability, the team can enhance the device's capability to withstand the demanding conditions faced by firefighters during

emergency operations. The careful consideration of these mechanical elements aligns with the goal of providing a robust and dependable tracking solution for use in real-world firefighting scenarios.

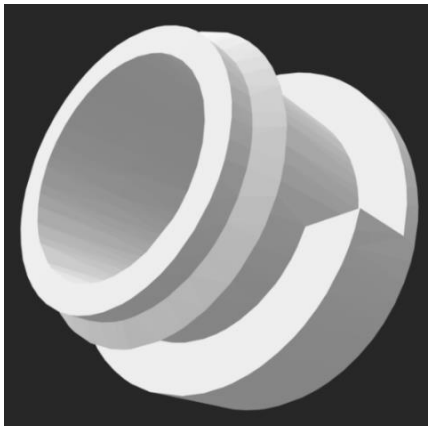


Figure 16: Casing Gasket Design [9]

Addressing the operational challenges posed by the thick and heavy gear, including gloves, worn by firefighters on the scene is a critical aspect of the device design. To facilitate effective operation in such conditions, the team plans to implement a large, arcade-style button, as illustrated in Figure 17. This button design is specifically tailored to meet the unique needs of firefighters who may be using the device with large gloves on or when the device is within their clothing.

The large, arcade-style button offers several key features that enhance usability in challenging environments:

- **Size and Ergonomics:** The button is designed to be sufficiently large, ensuring ease of use even with thick gloves. The arcade-style configuration provides a tactile and easily identifiable target for users.
- **Tactile Response:** The button is engineered to provide a tactile response upon being clicked. This feature is crucial for firefighters, as it allows them to feel the button's actuation, confirming that their input has been registered. This is especially important when operating the device in conditions where visibility may be limited.
- **Robustness:** The button design considers the robustness required for firefighting scenarios. It is crafted to withstand the demands of the environment, providing durability and reliability during critical operations.

By incorporating a large, arcade-style button with tactile feedback, the team ensures that the tracking device remains user-friendly and responsive in the challenging conditions faced by firefighters. This thoughtful approach to button design aligns with the overarching goal of

delivering a tracking solution that seamlessly integrates into the operational workflow of firefighters during emergency responses.

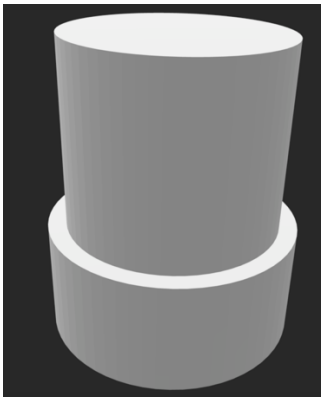


Figure 17: Large Arcade Button Design

The placement of the large arcade-style button is a crucial consideration to ensure ease of use and accessibility for firefighters in the field. While the specific location is not explicitly defined, it is essential to integrate the button in a manner that does not impede upon other functionalities of the device and allows for intuitive operation.

In Figure 18, the example showcases a potential integration point for the button within the casing design. This placement is designed to provide ergonomic access for firefighters, allowing them to interact with the button without hindrance. The goal is to position the button in a way that aligns with natural hand movements and ensures that users can operate it seamlessly, even when wearing gloves or in challenging conditions.

Adjusting the case design to accommodate the large arcade-style button is a key step in optimizing the device for practical use by firefighters. The thoughtful integration of the button into the overall

design reflects the team's commitment to creating a user-friendly and effective tracking solution tailored to the needs of emergency responders.

Figure 18: Theoretical Case Design with Button [9]

The consultation with the Livingston Fire Department (LFD) provided valuable insights into the preferred location for implementing the tracking device on firefighters. According to feedback from LFD firefighters, integrating the device onto their air packs is considered the most desirable option. This preference stems from the fact that air packs already house various meters and electronic devices, as depicted in Figure 19. When the air pack is turned on, it activates all connected devices simultaneously, streamlining the operational workflow for firefighters.

Figure 20 illustrates a proposed installation location on the strap of the air pack. Placing the device on the strap offers several advantages:

- **Convenience:** The strap location is easily accessible to firefighters. They can conveniently reach towards the shoulder straps to interact with the device, particularly when pressing the large arcade-style button to indicate their status.
- **Integration with Existing Gear:** Installing the device on the air pack aligns with the existing gear worn by firefighters. This integration minimizes the need for additional steps

or adjustments during emergency responses, ensuring a seamless and efficient user experience.

- **Operational Efficiency:** The device becomes an integral part of the air pack system, turning on and off in sync with the air pack. This synchronization simplifies the activation process for the tracking device, contributing to operational efficiency.

By aligning the device's location with the preferences of LFD firefighters and integrating it onto the air pack, the team ensures that the tracking solution seamlessly fits into the established workflows of emergency responders. This user-centric approach enhances the device's practicality and usability during critical situations.

Figure 19: Air Pack with Attached Electronic Devices

Figure 20: Potential device placement on air pack strap

The consideration of the device's exposure to the elements and the need for a heat-resistant polymer introduces an additional layer of complexity to the design. The suggestion from the Livingston

Fire Department (LFD) to explore efficient locations within the firefighter's clothing provides a viable alternative. Figure 21 illustrates potential insertion points, such as pockets in overalls or jackets, where the device can be securely placed without impeding the firefighters on the scene.

In this scenario, the choice of polymer may prioritize other factors such as durability and resilience over heat resistance. The emphasis shifts towards the critical aspects of the button design, considering the challenges posed by thick clothing and gloves. Key considerations for the button design include:

- **Surface Area:** The button should have a large surface area to facilitate easy operation through thick clothing and gloves. A larger button surface enhances tactile feedback, allowing firefighters to press it confidently.
- **Tactile Response:** Maximizing tactile response is crucial to ensure that firefighters can feel the button's actuation through their clothing and gloves. This feature is integral for confirming successful button presses, especially in situations where visibility may be limited.

As the team moves towards implementation, testing within the firefighter's clothing will provide valuable insights into the optimal location. The team's inclination towards device insertion within the clothing aligns with the practical considerations highlighted by the LFD. This user-centric approach ensures that the tracking device is seamlessly integrated into the firefighting gear, enhancing usability and overall effectiveness during emergency responses.

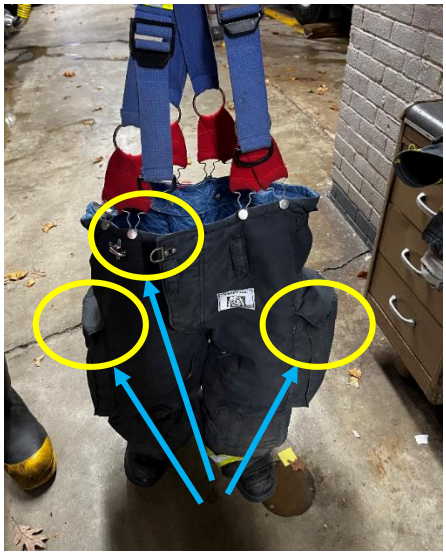


Figure 21: Potential device placement on firefighter overalls

Prototype Risk Assessment

The build phase of the project demands a thorough consideration of potential risks that could impact the procurement of a prototype. The primary risk identified is related to the attainment of materials, particularly with some parts having long lead times. To mitigate this risk, the team has proactively ordered and obtained the necessary materials for the upcoming semester.

Additionally, the team will continue to plan weeks in advance to ensure timely procurement of parts. In the event of unforeseen delays, the team is prepared to improvise where possible.

Secondary risks, such as potential disruptions in asset management due to team members missing class or falling ill, have been addressed with a contingency plan. If a team member assigned to a major subsystem is unable to fulfill their responsibilities, the team lead will reassign work accordingly. Other team members with less workload will be reassigned to compensate for the absence, with additional support provided by the remaining team members.

Given the short lifetime of the project, the team acknowledges the need to focus on a proof of concept for the prototype. However, for the system to be robust and complete, it is imperative that GPS and status indicator data be sent to a server to prevent data loss in the event of node damage. Any expected delays in the procurement of the prototype will be communicated directly with the sponsor to ensure transparency and alignment of expectations.

By proactively addressing these identified risks and implementing contingency plans, the team demonstrates a commitment to managing potential challenges during the build phase, ultimately contributing to the successful development of the tracking system prototype.

Chapter 4: Build

The team was divided into two sub teams: hardware and software. The hardware team was comprised of Rylee Butler, Michael Manning, and Phil Rodrigues. The software team was comprised of Jingyi Yen and Jack Torrence.

Software

The software portion of the project includes two aspects: coding the GUI, and harvesting the data sent from nodes using the Meshtastic based open-source software. The Python files and image files for icons can be found under the "Code Repo" link in Appendix B.

The GUI was coded in Python using the tkinter Python library. The GUI is split into two portions, the "main menu" and the "map view". The main menu is shown in Figure 22, with the map view in Figure 23.

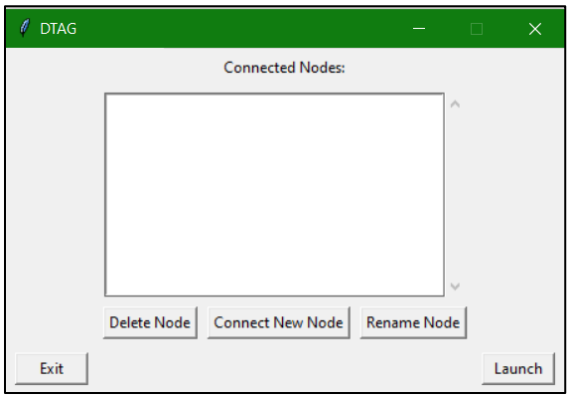


Figure 22: GUI "Main Menu"

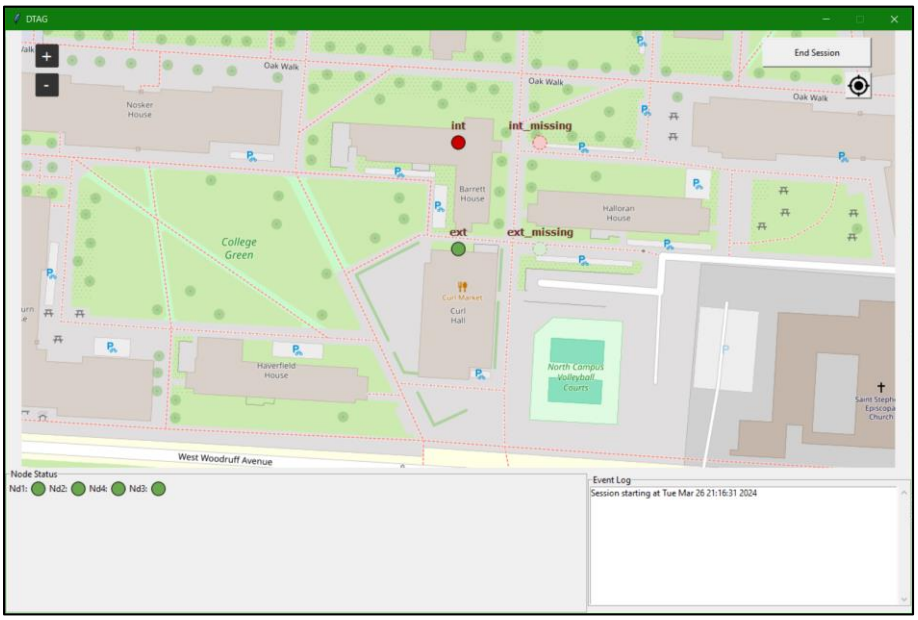


Figure 23: GUI "Map View"

The main menu is where the user can connect new nodes, rename them, and delete any nodes if desired. The map view is where the user can see the status of each node, an event log of any status changes, and the GPS location of each node on the map. The user can click and drag to move the map, zoom in and out with the "+" and "-" buttons. Finally, the user can press "End Session" to close the map and generate a log of the event log in the form of a word document.

The node information is harvested using Meshtastic Python and CLI (command line interface) commands initiated from our own generate Python script. Once "Connect New Node" in the main menu is pressed, a Python command is called to gather the information of each Meshtastic node that is in range of the node connected via USB to the laptop on which the GUI is running. The Python command returns a large string with all node information, in which our script parses through it and for each node saves an array of 10 items called the Nodes data structure as shown in Figure 24.

```
# current node data structure:
# [0]: node ID, str
# [1]: long name, str
# [2]: short name, str
# [3]: on or off, boolean
# [4]: node status, int: 0-3
# [5]: when button press for specific node: time, int
# [6]: last recieved any message: time, int
# [7]: lat, float
# [8]: long, float
# [9]: last time gps updated: time, int
currentNode = [nodeIDstr, longStr, shortStr, isOn, nodeStatus, 0, time.time(), lat, long, 0]
```

Figure 24: Nodes data structure.

While the map view is running, different packets of information are sent between each node. When a packet is received by the central gateway (node connected to laptop running GUI), an interrupt named onRecieve() is called. There are three different types of packets, Regular Pings sent every 60 seconds, GPS Info, and Button Press packet. On receiving a GPS packet, the packet is changed into a string, and the node ID, latitude, and longitude information is parsed and collected using the updateGPS() function. On a button press packet, the packet is changed into a string, and the node ID, time difference between presses for the same node is parsed and gathered. The GUI is then updated to reflect any positional or status changes with updateMarkers() and updateStatus(). This is illustrated in Figure 25. The rate of which the regular ping and GPS info packet is sent is configurable via the Meshtastic Web Client website.

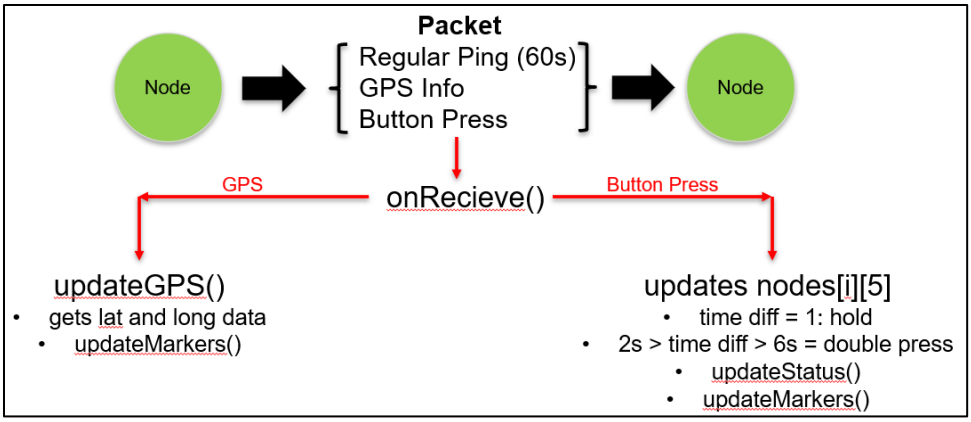


Figure 25: onRecieve() function high level breakdown

There is also an interrupt that happens every 10 seconds called checkTime(). This function checks for two things, how long ago any signal was received from each node to the central gateway, and how long a GPS signal was received from each node to the central gateway. If no signal was received from a certain node after 90 seconds, then the node is assumed to be off or unable to send any data, and the status of that node will be changed to "no signal" with the marker in the GUI placed at the last known location and with the last known type of status (interior or exterior). If no GPS signal was received from a certain node after 10 seconds, but messages to the central gateway are still able to be received, then the node will be changed to the "missing" status with the marker in the GUI placed at the last known location and with the last known type of status. The different status GUI symbols are shown in Figure 26.

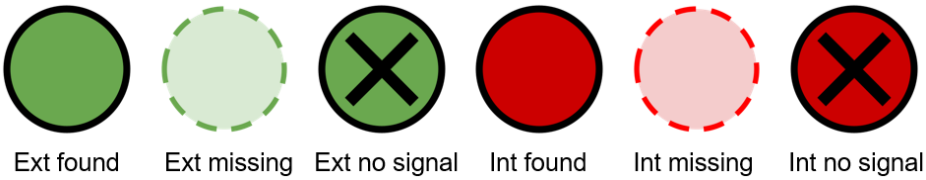


Figure 26: Different GUI status symbols

For the button presses to be registered on each node, remote hardware must be enabled for each node using Meshtastic CLI commands. Additionally, the channel URL of the central gateway node is registered to each node individually, allowing for the central gateway to send GPIO commands to each node. This is used so after a status is changed by using the buttons on each node, the central gateway can set GPIO pin 13 on the node to be 3.3V to play the speaker as a form of feedback that the central gateway successfully received the status change.

Hardware

The hardware portion of the project includes two circuits: the button circuit and the buzzer circuit. Throughout the build phase, the type of button and number of buttons changed. The button circuit is simple, with two buttons connected in series to ensure an intentional button press, as shown in Figure 27.

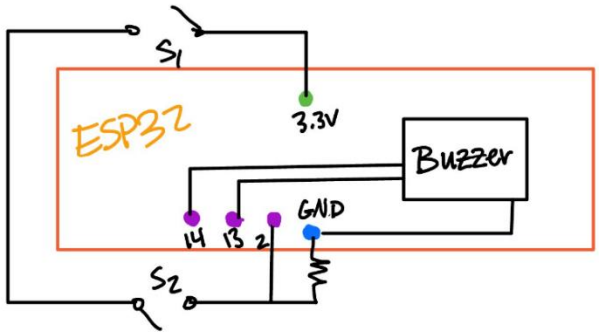
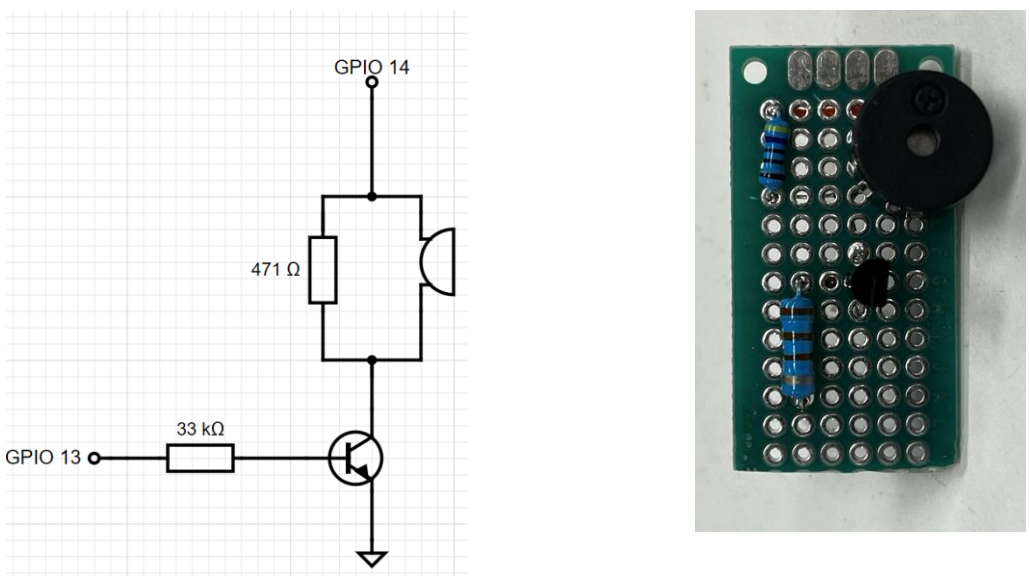


Figure 27: Circuit diagram of button connections and connections to buzzer circuit.

The team designed a buzzer circuit using two resistors, one NPN transistor, and two 3.3 V inputs provided by two GPIO pins on ESP32 board, as shown in Figure 28a. As there is only one 3.3 V output provided on the ESP32 board, the team opted to set GPIO 14 to HIGH and use GPIO 13 to toggle the buzzer on and off. When 3.3 V is applied to the 33 kΩ resistor, the transistor enters saturation and completes the circuit, sounding the buzzer. After completing the circuit on a breadboard, the team connected each external voltage input to a power supply set to 3.3 V to test the circuit functionality. After functionality was confirmed, the team continued to build the buzzer circuit using a perforated PCB, as shown in Figure 28b.



(a) (b)

Figure 28: (a) Buzzer circuit with GPIO pins 13 and 14 connected, GPIO pin 14 always set to HIGH, which outputs 3.3 V. GPIO pin 13 toggles the buzzer (b) Completed buzzer PCB circuiton and off.

After the completion of the button and buzzer circuits, the team used a breadboard to verify the functionality of the two circuits together, as shown in Figure 29. Finally, the hardware team

soldered the connections to all components to develop the final product. Casing was added to one of the nodes to display the button and buzzer placement envisioned during the design phase.

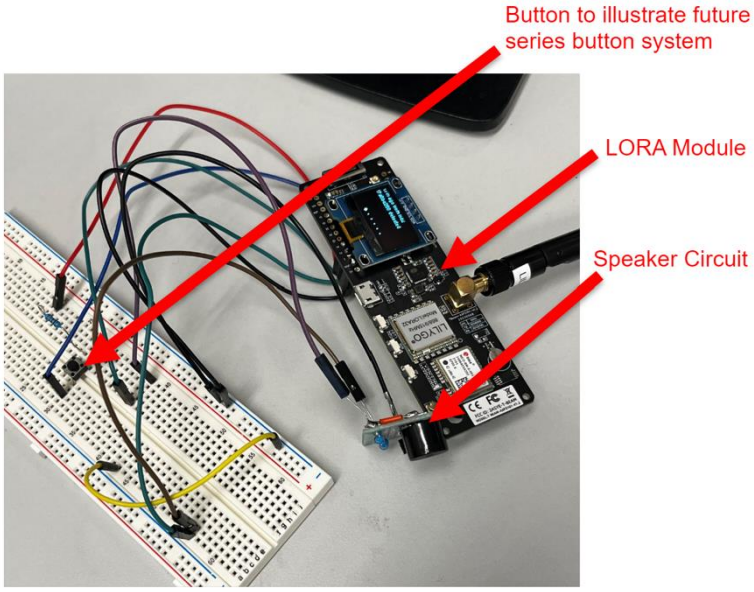


Figure 29: Button and buzzer circuits connected to ESP32 via breadboard to verify functionality.

Chapter 5: Test Plan and Results

Function Tests

Test 1.1: Tracking Status (Appendix C (Table C.1))

After completing the GUI build and preliminary programming of the ESP32 nodes, the team performed the tracking status test. The node was moved 50 meters away from the central gateway and tracked on the GUI. On the first test, the node's position appeared on the screen along with its status. Therefore, the test passed since the system worked as expected.

Test 1.2: Update Status of Trackers (Appendix C (Table C.2))

As an extension of Test 1.1, this test aimed to test whether the status of the node would change as it moved around its environment. So, after performing the Tracking Status test, the team performed the update status of trackers test. Again, the node was taken 50 meters away from the central gateway. This time, a team member walked around with the node as another observed the motion on the GUI. The team found that the tracking status was updated about every 5 seconds as desired. This test passed.

Test 1.3: Status Change (Appendix C (Table C.3))

After completing the build for the preliminary button circuit, the team performed the status change test. Using a breadboard circuit to simulate the button that was yet to be completed, the team ensured the button press would communicate a status change to the central gateway and mesh network. The test passed on the first attempt. The team also performed the test after soldering the finished button circuit to the node. The team programmed the buttons so that they would only respond to a long press. With a short press, the status should not change. After testing, the system worked as expected, so the team officially passed the test.

Test 1.4: Notification of Status Change (Appendix C (Table C.4))

Once the team completed the PCB circuit for the buzzer, the team performed the notification of status change test. Using a breadboard circuit to simulate the button that was yet to be completed, the team ensured the buzzer was triggered when the button was pressed. The test passed on the first attempt.

Test 1.5: Data Transmission (Appendix C (Table C.5))

Once the team had the nodes communicating the mesh network, the team performed the data transmission test. This test aimed to ensure the central gateway recognized the button press of another node. This data transmission would be confirmed by checking the central gateway after a button press to make sure the triggered node was detected (the screen would name the node and specifically say it was detected). After performing the test, the system worked as expected, so it passed.

Performance Tests

Test 2.1: Update Location Consistently (Appendix C (Table C.6))

This test simply checks to ensure the location is being accurately sent back to the GUI. A team member carried the node in a straight line while another team member watched the GUI to make sure the movement showed on the screen. This test was successful, so it passed.

Test 2.2: Battery Life (Appendix C (Table C.7))

After the hardware construction was completed, the team performed the battery life test. Before the test, the battery was fully charged. The battery was then placed back in the node, and the node was left on for eight hours. When the team checked the battery after hours, the battery dropped to 81% which far surpasses the requirement of eight hours, meaning the test passed on the first attempt.

Test 2.3: GPS Resolution (Appendix C (Table C.8))

To test GPS resolution, the team again moved around with the node at various distances from the central gateway to observe the error between the node's true location and the location displayed on the GUI. The team wanted the location of the node to be within 5 meters of the actual location at any given time. After testing and taking multiple data points, the team decided that it was successful since the location of the node on the GUI was always within 5 meters of the actual location. Therefore, this test passed.

Test 2.4: GPS Precision (Appendix C (Table C.9))

As an extension to Test 2.3, the team tested to see whether the distance in which the node moved on the GUI was the same as the actual distance the node moved in real-time. This includes the correct direction, distance, and time of movement. This test was performed simultaneously with Test 2.3 using the same data points, and it was determined the GUI was correctly and precisely updating the location of the node. Therefore, the test passed.

Test 2.5: Data Storage (Appendix C (Table C.10))

After ensuring nodes could communicate their status changes to the GUI, the team tested to see whether this information could be stored in a central file after an event. To do this, the team performed status changes on the nodes. After the session ended, the team checked to see whether the data was saved to the desired location. After this, the team checked to make sure the stored data matched the changes that occurred. These changes were stored in a Word document. The team determined the data storage was successful, so the test passed.

Test 2.6: Range (Appendix C (Table C.11))

The team walked at various distances away from the central gateway and other nodes to ensure the nodes still communicated with each other consistently at all distances. Again, one team member walked with the node while another stayed with the central gateway and observed the status on the GUI. The requirement was that the nodes communicated at 500 feet, but the team found that at 1000 feet, the nodes still communicated their status accurately. Therefore, the test was successful, and it passed.

Safety Tests

Test 3.1: Node Ruggedness (Appendix C (Table C.12))

The team decided that this test went beyond the scope of the project because of the limited time and resources available to create adequate housing for the nodes. This is something that should be addressed in future developments to this system.

Test 3.2: Node Physical Characteristics (Appendix C (Table C.13))

The requirement for each finalized node was for it to be less than 1 pound. The team weighed the node after the hardware team completed the build, and the node weighed 6 ounces. This clearly passed the test.

Test 3.3: Central Gateway Physical Characteristics (Appendix C (Table C.14))

The requirement for the central gateway including the computer was for it to be less than 10 pounds. The team weighed the central gateway and its components after the hardware team completed the build, and the node was 4.5 ounces. This clearly passed the test.

Other Tests

Test 4.1: Field Testing (Appendix C (Table C.15))

The team decided this test went beyond the scope of the project because of the limited time and resources available to perform full scale field testing. This is something that should be addressed in future developments to this system.

Test 4.2: Interference Testing (Appendix C (Table C.16))

The team tested the frequency band of the nodes using an oscilloscope in the Capstone lab and spectrum analyzer. In each test, the frequency band was between 902 MHz and 928 MHz which passed the test.

Test 4.3: Environmental Compatibility (Appendix C (Table C.17))

The team decided this test went beyond the scope of the project because of the limited time and resources available to create the adequate housing necessary to conduct an environmental compatibility test. This is something that should be addressed in future developments to this system.

Chapter 6: Redesign and Lesson Learned

Hardware Redesign

In the team's original design plans, there were detailed considerations regarding what the team would use regarding the casing for the module's components. But, given the scope of the project as an electrical engineering project, the team decided to use a simple plastic casing design that can be seen in Figure 26. The team decided to dedicate the casing development to a team of mechanical engineers in the near future when considering final product design, but in regards to the final prototype, the current casing design suffices.

Figure 26: Prototype design with full casing

Additionally, in the original design the team had planned to utilize the Lora 32 Module's OLED screen to notify users of device status changes. But, during the design process, the team decided that this design feature would not be worth the additional work and challenges that came with it, as the devices would be within the interior of the user's gear anyways, therefore an OLED screen notification would prove essentially useless in notifying the user. In this regard, the team decided to focus on the speaker and using sound to notify the user of status changes, and the OLED screen design unanimously fell out of favor in the development process.

The design of the Module and the Speaker and Button circuits remained largely unchanged from the original detailed design and build plans, and the team was able to successfully create what it had sought to develop in this regard. One minor difference in the design plan was adding a second button in series with the original button in circuit. This is because the team had decided later in the design process to integrate a fail-safe in the button design, requiring both buttons to be simultaneously pushed down in order to initiate a status change, as just one button could easily be

accidentally pressed down within a user's gear. One other minor change was that the speaker circuit utilizes 2 GPIO pins on the Lora module as opposed to 1, given that the transistor circuit requires two 3.3V power sources in order to effectively toggle between on and off states.

In regards to software redesign, there is not much to be addressed as the software design was a continuous process which almost completely went to plan for this team.

Lessons Learned

Overall, the team learned that sometimes redesign for projects is necessary to deliver an operational project that meets both timeline and function objectives. In this case, while the team had higher aspirations regarding "nice-to-haves" in the design process such as the casing and OLED screen, redesign changes were necessary in order to meet timeline restrictions. In real-world projects, timelines are going to be set, and there are going to be project "must-haves" and project "nice-to-haves". The team believes that this capstone project had many parallels to that of a real-world industry project, and taught the team the importance of efficient and effective redesign decisions in order to meet project goals. While the team had aspired to do more, without effective redesign decisions the team would currently be with a non-operational final prototype. For this reason, the team dedicates much of it's success to that of the redesign and development, as it indirectly led to the team's final functional prototype.

Chapter 7: User Manual

After starting the code, the GUI will enter its initial state. The GUI will show a list of currently connected nodes, and a set of buttons. The buttons are “Delete Node”, “Connect New Node”, “Rename Node”, and “Launch”. Clicking, “Connect New Node” will display a list of the nodes in the area. After clicking one of these nodes, the “Rename Node” and “Delete Node” buttons can be used. The “Rename Node” button brings up a pop-up allowing the user to enter a new name for the node. The “Delete Node" button deletes the node from the list. The “Connect New Node” button can be used to re-add deleted nodes to the list. Finally, the “Launch” button will start the tracking of nodes status and location.

Clicking the “Launch” button will bring up the Main Window. This window displays the “Event Log”, a history of node status changes, the “Node Status” window, which shows the current status of each node, and a map. There is also a button labeled “End Session”. The map starts off centered at the location of the central gateway and shows the location of each node as well as it’s status. There are six statuses, each with a corresponding symbol. A node in the exterior that is able to send GPS and status information will be represented as an opaque green circle with a solid black outline. A node in the exterior that is unable to send GPS information, but able to send status information, will be represented as translucent green circle with a dashed green outline. A node in the exterior that is unable to send GPS or status information will be represented as a solid green circle with a solid black outline and a black “X” through it. This pattern is repeated with nodes in the interior except instead of green, they are red.

A node must be turned on for it to be tracked. To turn on a node simply hold the left most black button until the screen turns on. The node will automatically broadcast GPS and status information. To change the status (exterior to interior) or vice versa. The two red buttons must be pressed at the same time. To change from exterior to interior, hold down both buttons for 3-4 seconds. To change status from interior to exterior, press both buttons at the same time, wait 2-4 seconds and the press both buttons again. In a few seconds, the speaker on the node will sound. This speaker is confirmation that the change in status has reached and was recorded at the central gateway. If the speaker does not sound, the status change did not reach the central gateway.

At the end of a session simply click the button labeled “End Session” and confirm the end of the session. This will close the GUI. The event log of the session will be saved as a Word Document that is titled the date and time the session was started.

Chapter 8: Project Management

The team has established a team charter to ensure efficient collaboration, outlining key expectations and communication tools. Regular meetings, Microsoft Teams, and accompanying software have been designated as primary communication channels. Research tools include Meshtastic, an open-source mesh network, Discord Server, and YouTube Channel. MATLAB will be used for verification and simulation, and Android Studio for UI development if the precise mapping system concept is pursued.

A Gantt Chart has been devised to coordinate design, build, and testing phases, with team members assigned tasks based on their roles and expertise. Jack and Jingyi will program the ESP32 nodes and the Meshtastic Web Client while Michael and Phil will lead the hardware design and construction of the ESP32 with the external functional components. Rylee, as the team lead, will act as the bridge between the software and hardware teams to ensure the team remains on schedule. The estimated timeline involves finalizing the design by December, material procurement over the Christmas break, and prototype construction in January. However, the schedule is subject to change based on deliverable due dates and project complexity.

Identified risks include uncertainties in tracking resolution and potential signal issues between nodes and the main gateway, especially in complex environments. Limited initial familiarity with the technology poses a risk due to the project's short lifetime.

High-level goals involve addressing and improving the current personal accountability system used by firefighters. The aim is to automate and enhance the safety processes by implementing an efficient and effective automated personal accountability system. Technically, the team aims to utilize LoRa technology for tracking first responders' general location in emergencies. Continuous communication with firefighters, including Fire Chief Mr. Wortman and Firefighter Lucas Giordano, is prioritized to align the project with the needs of first responders and ensure accurate development of the solution.

Appendix A

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Appendix B

Team Charter

Meeting Notes

Weekly Progress Reports

Interview with Adamsville Volunteer Fire Dep. Fire Chief

Research

Gantt Chart

Problem Identification PowerPoint Presentation

Preliminary Design Review PowerPoint Presentation

Critical Design Review PowerPoint Presentation

Final PowerPoint Presentation

Parts List

Code

Appendix C

Table C.1: Tracking status test

1.1 Tracking Status Test	
Method of Verification: Demonstration	
Type of Test: Integration	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop (x1), Lora ESP32 (x2)	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – Central console GUI displays location and status of node.** • **Fails – Central console GUI does not display location or status of node.**	
Test Procedure: 1. Open Python based DTAG GUI on laptop. 2. Connect one Lora ESP32 board to be used as the Central Gateway via Bluetooth to the laptop. 3. Connect and assign the other ESP32 board via Bluetooth as a node. 4. Move the node at least 50 meters away from the Central Gateway. 5. Change status on the node by pressing the status button on the node. 6. Passes test if the location and the different status of the node is shown and recorded on the Central console GUI, on the map and node status indicators respectively. Fails test if either the location or status of the node is not shown on the GUI. 7. If the test fails, consider moving the node to an open space, with no possible obstructions overhead for a clearer signal.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jingyi Yen	Witness: Michael, Jack

Table C.2: Update status of trackers test

1.2 Update Status of Trackers	
Method of Verification: Test	
Type of Test: Integration	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop (x1), Lora ESP32 (x2)	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – Central console GUI updates location of nodes every 5 seconds.** • **Fails – Central console GUI does not update location of nodes every 5 seconds.**	
Test Procedure: 1. Open Python based DTAG GUI on laptop. 2. Connect one Lora ESP32 board to be used as the Central Gateway via Bluetooth to the laptop. 3. Connect and assign the other ESP32 board via Bluetooth as a node. 4. Move the node at least 50 meters away from the Central Gateway. 5. Continuously move the node around while still maintain a GPS signal. 6. Passes test if the location of the node is shown and recorded on the Central console GUI every 5 seconds as the node moves around. Fails test if the location is not shown moving with the node on the GUI. 7. If the test fails, consider moving the node to an open space, with no possible obstructions overhead for a clearer signal.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jingyi Yen	Witness: Michael, Jack

Table C.3: Status change test

1.3 Status Change Test	
Method of Verification: Test	
Type of Test: Integration	
Date(s)/ Total Times: **3/23/2024**	**Location: Dreese Labs 761**
Test Equipment: Laptop, Lora ESP32 (x2)	
Test Engineer: Jack Torrence **Witness: Jingyi Yen**	
Criteria for Success: • **Passes – Status of node changes to reflect button presses** • **Fails – Status of node doesn't change to reflect button presses**	
Test Procedure: 1. Connect one Lora ESP32 board to be used as the Central Gateway to the laptop. 2. Connect and assign another ESP32 board as a transponder. 3. Press the buttons twice in quick succession (under 3 seconds between presses) 4. Check on laptop to ensure data where node status is stored shows interior 5. Press and hold buttons (over 2 seconds) 6. Check on laptop to ensure data where node status is stored shows exterior	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack Torrence	Witness: Jingyi Yen

Table C.4: Notification of status change test

1.4 Notification of Status Change	
Method of Verification: Demonstration	
Type of Test: Integration	
Date(s)/ Total Times: 4/3/2024	**Location: Dreese Labs 761**
Test Equipment: Laptop (x1), Lora ESP32 (x2)	
Test Engineer: Jack Torrence **Witness: Rylee Butler**	
Criteria for Success: • **Passes – Node successfully changes status on GUI. Speaker indicates Node has changed status.** • **Fails – Node does not change status on GUI or speaker do not indicate Node has changed status.**	
Test Procedure: 1. Open Python based DTAG GUI on laptop. 2. Connect one Lora ESP32 board to be used as the Central Gateway via Bluetooth to the laptop. 3. Connect and assign the other ESP32 board via Bluetooth as a node. 4. Change status on the node to interior by holding the status button on the node. 5. Change status on the node to exterior by pressing the status button on the node. 6. Passes test if the status on the GUI changes, and the speaker on the node rings a tone according to which status it is being changed to. Fails test if either the status on the GUI or the speaker do not change when the status changed. 7. If the test fails, consider restarting the nodes and the GUI.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack Torrence	Witness: Rylee Butler

Table C.5: Data transmission test

1.5 Data Transmission Test	
Method of Verification: Test	
Type of Test: Demonstration	
Date(s)/ Total Times: **3/23/2024**	**Location: Dreese Labs 761**
Test Equipment: Laptop, Lora ESP32 (x2)	
Test Engineer: Jack Torrence **Witness: Jingyi Yen**	
Criteria for Success: • **Passes – Button press is recognized at central gateway** • **Fails – Button press is not recognized at central gateway**	
Test Procedure: 1. Connect one Lora ESP32 board to be used as the Central Gateway to the laptop. 2. Connect and assign another ESP32 board as a transponder. 3. Press button on transponder 4. Ensure central gateway recognizes button press by checking on the laptop	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack Torrence	Witness: Jingyi Yen

Table C.6: Location consistency test

2.1 Location Consistency Test	
Method of Verification: Test	
Type of Test: Integration	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop, Lora ESP32 (x2)	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – Location displayed on computer reflects movement of transponder node** • **Fails – Location displayed on computer doesn't reflects movement of transponder node**	
Test Procedure: 1. Connect one Lora ESP32 board to be used as the Central Gateway to the laptop. 2. Connect and assign another ESP32 board as a transponder. 3. Carry transponder node in straight line in an area with no overhead obstructions 4. Ensure location of node displayed on laptop matches the path of the transponder node with waypoints made every 5 seconds	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack	Witness: Jingyi

Table C.7: Battery life test

2.2 Battery Life Test	
Method of Verification: Test	
Type of Test: Component	
Date(s)/ Total Times: **4/10/2024**	**Location: Off Campus (Rylee's House)**
Test Equipment: Laptop, Lora ESP32 (x2)	
Test Engineer: Rylee Butler **Witness: Jack Torrence**	
Criteria for Success: • **Passes – Battery successfully operates device in session for at least 8 hours time** • **Fails – Battery operates device in session for less than 8 hours before dying**	
Test Procedure: 1. Begin session and add node to be tested 2. Check to confirm a full battery charge 3. Run session so that node receiving and sending data begins operating at full intensity 4. Set an 8-hour timer for this live session to run. 5. If timer runs out before battery dies, battery life will be deemed efficient to sustain activity of a full session.	
Action Items: **If the test failed what needs to be done and determine the retest date and location** Battery drained 19% over a span of 8 hours (battery life was at 100% at beginning of test and at 81% at end of test)	
Test Engineer: Rylee Butler	Witness: Jack Torrence

Table C.8: GPS resolution test

2.3 GPS Resolution Test	
Method of Verification: Analysis	
Type of Test: Component	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop, Lora ESP32 (x2), Antenna	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – Location of user on GUI is consistently located within 5 meters of user's true location** • **Fails – Location on GUI is not consistently or not at all within 5 meters of user's true location**	
Test Procedure: 1. Have an engineer stand with 1x Lora ESP32 module in an easy to accurately mark location on GUI world map 2. Begin a tracking session on laptop 3. Set up antenna with central gateway/laptop 4. Connect testing node being held by engineer 5. Measure distance accuracy of displayed GUI icon for engineer's LORA ESP32 location 6. If distance is within 5 meters of true location, engineer will then be asked to walk for approximately 50 meters 7. Check to see if GUI icon continues to maintain at least 5 meter distance from device's true location. 8. If maintains at least 5 meter distance, resolution of GPS will be deemed satisfactory.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack	Witness: Jingyi

Table C.9: GPS precision test

2.4 GPS Precision Test	
Method of Verification: Analysis	
Type of Test: Component	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop, Lora ESP32 (x2), Antenna	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – GUI Icon consistently moves same distance on map as the real device truly covers and in accurate direction** • **Fails – GUI Icon does not precisely move the same distance as is truly covered by the device or not in the proper direction**	
Test Procedure: 1. Have an engineer stand with 1x Lora ESP32 module in an easy to accurately mark location on GUI world map 2. Begin a tracking session on laptop 3. Set up antenna with central gateway/laptop 4. Connect testing node being held by engineer 5. Have engineer walk a measured distance of 5 meters 6. Measure change in distance of GUI icon on world map from previous location 7. If distance is approximately 5 meters (give or take 1 meter), then check the direction of updated icon location with compass 8. If direction of change is the same as true device's direction of change, then GPS precision will be deemed satisfactory.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack	Witness: Jingyi

Table C.10: Data storage test

2.5 Data Storage Test	
Method of Verification: Inspection	
Type of Test: Integration	
Date(s)/ Total Times: **3/23/2024**	**Location: Dreese Labs 761**
Test Equipment: Laptop, Lora ESP32 (x2)	
Test Engineer: Jack Torrence **Witness: Jingyi Yen**	
Criteria for Success: • **Passes – Data saved for report matches inputs received during session** • **Fails – Data saved for report doesn't match inputs received during session**	
Test Procedure: 1. Connect one Lora ESP32 board to be used as the Central Gateway via to the laptop. 2. Connect and assign another ESP32 board as a transponder. 3. Make status and location changes with the transponder 4. Record these changes as they happen on the laptop 5. End session 6. Ensure data saved after sessions ends matches the changes recorded earlier	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jack Torrence	Witness: Jingyi Yen

Table C.11: Range test

2.6 Range Test	
Method of Verification: Test	
Type of Test: Integration	
Date(s)/ Total Times: **3/26/2024**	**Location: Annie and John Glenn Ave, Columbus, OH**
Test Equipment: Laptop, Lora ESP32 (x2), Antenna	
Test Engineer: Jingyi Yen **Witness: Jack Torrence**	
Criteria for Success: • **Passes – Tracker module successfully sends data while within 500 feet of central gateway and/or any other connected tracker** • **Fails – Tracker module unsuccessfully sends data while within any area 500 feet from the central gateway and/or any other connected tracker**	
Test Procedure: 1. Have an engineer stand with 1x Lora ESP32 module. 2. Begin a tracking session on laptop 3. Set up antenna with central gateway/laptop 4. Connect testing node being held by engineer 5. Have engineer walk away from central gateway/laptop for a distance of 500 feet 6. See if GUI icon consistently updates location and status data for the duration of 500 feet walked 7. If step 6 is completed successfully, then connect second tracker node to session 8. Have second tracker positioned 500 feet away from central gateway 9. Then, have engineer walk first tracker 1000 feet away from central gateway in exact same direction as the second tracker 10. Check to see that first tracker GUI icon consistently updates location and status data for the duration of 1000 feet walked 11. If step 10 is successful, range ability requirements for the tracker nodes will be deemed satisfactory.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Jingyi Yen	Witness: Jack Torrence

Table C.12: Node ruggedness test

3.1 Node Ruggedness	
Method of Verification: Test	
Type of Test: Component	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Lora ESP32 (x1)	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – There is no apparent damage to the ESP32, and it is fully functional** • **Fails – There is apparent damage to the ESP32 and/or it is not fully functional**	
Test Procedure: 1. Place the Lora ESP32, with case attached, on the ground. 2. Place a 30-lb weight (~135 N) on top of the Lora ESP32. 3. Remove the weight and examine the node for damage. Test for functionality.	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer:	Witness:

Table C.13: Node physical characteristics test

3.2 Node Physical Characteristics	
Method of Verification: Test	
Type of Test: Component	
Date(s)/ Total Times: 4/3/2024	**Location: Dreese Labs 761**
Test Equipment: Lora ESP32 (x1), scale (measure in lbs)	
Test Engineer: Rylee Butler **Witness: Phil Rodrigues**	
Criteria for Success: • **Passes – Node weighs less than or equal to 1 lbs** • **Fails – Node weighs more than 1 lbs**	
Test Procedure: 1. Place one Lora ESP32 node on scale with case attached. 2. Record weight reading on the scale.	
Action Items: **If the test failed what needs to be done and determine the retest date and location** **Weight: 6.4 oz.**	
Test Engineer: Rylee Butler	Witness: Phil Rodrigues

Table C.14: Central gateway physical characteristics test

3.3 Central Gateway Physical Characteristics	
Method of Verification: Test	
Type of Test: Component	
Date(s)/ Total Times: **4/11/2024**	**Location: Dresse Labs 761**
Test Equipment: Laptop (x1), Lora ESP32 (x1)	
Test Engineer: Rylee Butler **Witness: Michael Mannning**	
Criteria for Success: • **Passes – Weight is less than or equal to 10 lbs.** • **Fails – Weight is greater than 10 lbs.**	
Test Procedure: 1. Place one Lora ESP32 node and one laptop on a scale with case attached. 2. Record weight reading on the scale.	
Action Items: **If the test failed what needs to be done and determine the retest date and location** 4lbs 7.5 oz	
Test Engineer: Rylee Butler	Witness: Michael Manning

Table C.15: Field test

4.1 Field Testing	
Method of Verification: Demonstration	
Type of Test: System	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Laptop (x1), Lora ESP32 (x2)	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – Firefighter can change the status with full gear including gloves** • **Fails – Firefighter cannot change status with full gear including gloves**	
Test Procedure: 1. Equip one test subject with thick gloves similar to the thickness of a typical firefighter's gloves 2. Begin a session and add the node to be tested 3. Once the session has started, have the test subject press the buttons to change their status to inside the building 4. Have the test subject press the buttons to change their status to outside the building 5. Observe the status changes on the GUI to ensure the status is changing as desired	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer:	Witness:

Table C.16: Interference test

4.2 Interference Testing	
Method of Verification: Test	
Type of Test: Component, System	
Date(s)/ Total Times: 4/9/2024, 4/10/2024	**Location: Dreese Labs 761**
Test Equipment: Laptop (x1), Lora ESP32 (x2), Oscilloscope (x1)	
Test Engineer: Rylee Butler, Jingyi Yen **Witness: Jack Torrence**	
Criteria for Success: • **Passes – Frequency band of nodes is in the 915 MHz unlicensed frequency band (902-928 MHz)** • **Fails – Frequency band of nodes is outside the 915 MHz unlicensed frequency band (<902 or >928 MHz)**	
Test Procedure: 1. Turn on the ESP32 nodes 2. Use a spectrum analyzer to test the frequency band sent between the nodes 3. Observe the results and discuss alternatives if necessary	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer: Rylee Butler, Jingyi Yen	Witness: Jack Torrence

Table C.17: Environmental capability test

4.3 Environmental Capability	
Method of Verification: Inspection, Demonstration	
Type of Test: System	
Date(s)/ Total Times: N/A	**Location: N/A**
Test Equipment: Lora ESP32	
Test Engineer: XYZ **Witness: ABC**	
Criteria for Success: • **Passes – The ESP32 node continues to communicate and send proper status changes after getting splashed by water or covered in dust** • **Fails – The ESP32 node does not communicate or send proper status changes after getting splashed by water or covered in dust**	
Test Procedure: 1. Ensure the node is enclosed within its casing; the case should be closed without any gaps or bulging 2. Splash the casing with a handful of water 3. Press the button to initiate a status change 4. Observe whether the status change was communicated to the central gateway and displayed on the GUI	
Action Items: **If the test failed what needs to be done and determine the retest date and location**	
Test Engineer:	Witness: